

Eins System: A Scientific, Fair, Real-Time Measure of AI Originators’ Influence on Society

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Eins Arbitration Committee

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Abstract

This white paper presents the Empirical Influence Norm System (EINS), a research measurement system whose flagship output is the Eins Index (code 8100): a scientific, fair, real-time measure of the influence that *AI Originators* — the communities that continuously create, release, and iterate AI models and thereby supply AI productive capacity to society — exert on society. The index *output* layer is designed to be accessible: from a full indicator state, a small set of well-defined public artifacts (a single number, a single line, a consistent standard) is produced by an explicit presentation-layer mapping, so that a non-specialist can read the landscape without mastering the underlying measurement technology. Accessibility is a property of that output layer.

The scientific contribution of the system is a *measurement scale* rather than a single number. The scale is *relative* (influence is measured against a cross-sectional reference set rather than an arbitrary absolute), *centered* (a conservation law ties the sum of the index to the sum of its fundamental anchors, so that transient discussion reorders positions but cannot enlarge the total pool), and *real-time* (a collection–pricing–display pipeline publishes at minute level). The slow anchor is the *Fundamental Anchor* (FA): a composite of capability, edge-model coverage, ecosystem stock, and commercial accessibility. The live component is the *Real-time Influence* (RI) signal: an algebraically explicit renewal-envelope process whose numerical parameters are deliberately not disclosed.

The system is only partially reproducible and is designed to remain scrutable. Its invariants (conservation, centralization, boundedness, audit inequalities) are stated as propositions with proofs and are in principle checkable against public data, while the fast, easily manipulated components are kept opaque by design. FA methods and reference parameters are documented in this white paper; RI internals are reserved. This asymmetric disclosure is the institutional basis of the system’s trustworthiness.

Keywords: influence measurement; AI Originators; relative index; real-time index; measurement unit; construct validity; public understanding of AI.

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Part I

Motivation and Positioning

1 Introduction: Why Society Needs a Measurable Notion of AI Influence

1.1 A public cognitive gap

In 2026 the general public encounters a stream of names — GPT, DeepSeek, Claude, Gemini, Kimi, Qwen, Codex, Trae, Cursor — and a stream of benchmarks that purport to rank their “capability.” None of this tells a non-specialist where the field stands. Capability leaderboards describe *models*; product announcements describe *vendors*; neither describes the *structure of influence* — which communities are changing how society works, how fast, and in what direction. The vision behind this system is that AI concept and information equality is part of the public-good promise of AI itself: the same way a market index lets a layperson read the state of an economy, a well-built influence index should let a layperson read the state of the AI landscape. To the best of our knowledge from searchable public information, no public-facing index of *AI influence* existed for a general audience at the time of writing (Section 2.5); the index family presented here is an attempt to fill that gap with a measurement that remains defensible to scrutiny.

The gap is not merely aesthetic. A public that cannot distinguish “who is responsible for the AI changing my work” from “which vendor sent me the most emails” cannot form preferences over the direction of the field, cannot hold the relevant communities to account, and is structurally vulnerable to whichever narrative is loudest. Information equality about the AI landscape is therefore not a luxury layered on top of the technology; it is part of the social contract that the technology implies. An influence index is one instrument for providing it. It is not the only instrument, and this paper is careful (Sections 1.6 and 12.4) not to claim it is.

1.2 Capability is not influence

The central premise of this system is that *capability* and *influence* are different quantities, and that confusing them corrupts public reasoning about AI. We give three arguments, each with its most serious counterargument.

Argument 1: capability can be gamed. Whenever a quantity is used as a target, the statistical regularities behind it tend to collapse [Goodhart, 1984, Strathern, 1997, Campbell, 1979]; the phenomenon has been taxonomized and formalized for machine learning under labels such as Goodhart’s law variants [Manheim and Garrabrant, 2018], reward over-optimization [Gao et al., 2023], reward hacking [Skalse et al., 2022, Amodei et al., 2016, Pan et al., 2024], and benchmark data contamination [Balloccu et al., 2024, Koch et al., 2021]. A capability score is precisely such a target: it can be optimized by engineering effort in a way that need not track whether the models are *used*, *discussed*, and *built upon* by society. Measuring “capability” therefore measures a necessary input to influence, not influence itself. Section 2.2 assembles the empirical record of this failure mode at length, because it is the strongest argument for why a benchmark composite cannot substitute for an influence measure.

Counterargument and reply. One might reply that capability is the only “objective” quantity available, and that a noisy capability score beats an influence score whose very definition is contestable. The reply is twofold. First, the objectivity of a capability score is overstated: the empirical literature on benchmark contamination [Balloccu et al., 2024, Koch et al., 2021, Hendrycks et al., 2021]

and on machine-assessor bias [Zheng et al., 2023, Wang et al., 2023] shows that “objective” scores are themselves artifacts of an evaluation pipeline that can be and has been gamed. Second, the counterargument concedes the premise we deny — that a univariate capability ranking is the right object — and merely disagrees about noise. An influence measure is not a noisier capability score; it is a different quantity built from different inputs (use, discussion, iteration, diffusion), and it can be designed to be harder to game, as Section 7.5 argues.

Argument 2: omniscient and value-based assessments are value-contaminated. Any attempt to judge which models are “better for humanity” is forced to adopt a standpoint on what society should value, and different societies reach different standpoints; the resulting judgments inherit the values of their assessors [Winner, 1980, Douglas, 2004, 2009, Elliott, 2022]. Even a well-intentioned composite “quality” score quietly embeds a value choice. An influence measure should therefore minimize the amount of evaluation it performs and instead record *observable social fact*: who is used, by how much, by whom, and with what consequence. The empirical literature on LLM-as-a-judge confirms the practical bite of this worry: machine assessors exhibit systematic self-preference, position, and value biases [Zheng et al., 2023, Wang et al., 2023, Santurkar et al., 2023]. (We return to this theme in Section 7.2.)

Counterargument and reply. One might reply that “influence” is itself a value-laden construct — that choosing to weight usage over, say, safety is a value choice, so the system merely moves the problem. The reply is that the claim is narrower and testable. The system does not claim to be value-free; it claims that the set of value choices it must make is smaller and more transparent than the set a holistic quality judgment requires, and that each choice is designed to be auditable (Section 5.2 maps every weight to the influence definition). This is the distinction Douglas [Douglas, 2009] draws between values that enter a measurement as inescapable background and values that silently do the measuring: the design goal is to move as many choices as possible from the latter to the former. Whether it succeeds is an empirical claim about the audit trail, not a verbal claim.

Argument 3: value arises through use, discussion, iteration, and diffusion. Capability becomes social value only through a chain: a model is released; it is *used* in products and workflows; it is *discussed* publicly; it is *iterated* upon by developers and researchers; and it *diffuses* into edge devices, small models, and new applications. We therefore define the object of measurement as a composite of these four channels rather than as a single score of technical ability. This definition is operationalized in Section 5, given a construct-validity treatment in Section 4.2, and formalized in Section 6.

Counterargument and reply. One might reply that use without capability is impossible, so capability is a sufficient statistic after all. The reply is empirical: usage, discussion, and ecosystem stock are visibly not monotone functions of capability. A model that is adequate and free, that is embedded in a large developer ecosystem, or that is diffused through edge devices can generate more social value than a slightly more capable model locked behind a high price and a closed walled garden. Section 11 shows this structure in the first FA snapshot: capability leaders are not ecosystem leaders. The premise of “capability $\Rightarrow$ influence” fails as a measurement identity.

1.3 A measurable object: AI Originators

Because capability alone is not influence, and because influence flows from communities that produce and propagate models rather than from models in isolation, the system takes as its measurement unit the *AI Originator*: the community that continuously creates, releases, and iterates AI models and thereby supplies AI productive capacity to society. This is a class-level object — deliberately not a company, a brand, or a legal person (Section 3). The measurement question becomes: how much influence does each AI Originator exert on society, measured relative to the others, on a stable

scale, in something close to real time?

The choice of unit deserves scrutiny, because every index is hostage to its unit. Measure influence at the level of the *model* and you lose the organization that funds, hosts, and propagates it; measure at the level of the *company* and you miss open-source collectives, joint initiatives, and the boundary where a company and its research lab are not the same actor; measure at the level of the *country* and you average away the very communities whose behavior you want to observe (Section 2.1). The Originator class is written to admit all of these forms. We are honest about the current roster (Section 3.6): at present the included set is company-dominated, because the verifiable public sources we rely on are organized that way. The class definition is the more general object; the roster is a time-indexed instantiation of it.

1.4 Why an index

A single number, a single line, and a single consistent standard of measurement are not simplifications of the science; they are a *design choice about the presentation layer*. The design premise is that publics reason well with quantities they can track over time and compare across names. The index family presented here (Section 6) therefore behaves like a market-semantics quote machine for AI influence: a real-time sequence of values, periodic review-and-release of the slow anchor, and a suspend-resume release mechanism when measurement integrity is at risk. The semantics of a price quote are adopted; the financial products themselves are not, and the disclaimer in Section 13.6 makes this boundary explicit.

The “why an index” question has a scientific as well as a communication answer. A scale that can be summed (the flagship’s conservation law), compared across time (the settlement calendar), and disaggregated (the index family) is a scale designed to be *auditable*. A single published number is not a simplification that hides the science; it is the smallest object on which the science’s invariants can in principle be checked by an outsider. This is the sense in which the index form is load-bearing for trustworthiness, not merely for communication.

1.5 Contributions

The contribution is best stated at the level of the method system rather than of any single component: a *measurement unit* (the AI Originator, defined as a class), a *measurement scale* (relative, centered, real-time, with a conservation law at its center), and a *trust architecture* (asymmetric disclosure: open what is hard to manipulate, closed what is easy to manipulate). Individual methodological elements have precursors — model-level relative ratings, organizational capability scores, country-level AI indexes, citation-style influence measures — but we found no system combining an originator-set measurement unit with a centered, real-time, conservation-anchored scale (Section 2.5). Two properties deserve the modest framing of *design claims* rather than “firsts”: the system is, to our knowledge, the first public influence measure for AI Originators *as contributors*, and the first to give that influence a relative, centered, real-time scale.

A third contribution is methodological and runs through the whole paper: the system is presented as a *construct-validity case* rather than as a “dashboard.” Section 4 treats “influence on society” as a theoretical construct in the sense of Cronbach and Meehl [Cronbach and Meehl, 1955] and Messick [Messick, 1995], derives a content-valid operationalization from the definition, and states the convergent and discriminant evidence an observer would need before trusting the index. This is unusual for index engineering, and it is deliberate: an influence index that cannot say what would falsify it is a publication instrument, not a measurement.

1.6 Scope and non-goals

This system describes an assessment-method layer; it does not describe a data product, does not produce investment information, and does not constitute financial information, investment advice, a credit assessment, a reputation judgment, or a basis for individual decisions (Section 13.6). The system’s current measure is bounded by the sources it can verify; it reflects what has already happened, not a prediction of what will happen (Sections 8.9 and 12.4). Non-goals are stated as carefully as goals: the system does not adjudicate model quality, does not rank “best models for X,” does not predict future winners, and does not allocate responsibility in the governance sense (Section 2.3). What it does is narrower and, we argue, still hard: it assigns a defensible, auditable-by-design, real-time number to the question “how much social influence does each AI Originator currently exert, relative to the reference set?”

2 Prior Measurement Systems and the Positioning of This Work

2.1 An ontological map of measurement units

Prior measurement systems can be mapped onto two axes: the *measurement unit* (model, organization, company, country, or individual) and the *measured quantity* (capability, readiness, output, or influence). Table 1 lays the map out; it makes the positioning of the present work precise, and it makes the novelty claim checkable rather than rhetorical. The claim the table supports is not “no relative index exists” but “the cell (unit = AI Originator community, quantity = social influence, form = relative + centered + real-time) is empty.”

Table 1: An ontological map of measurement units (rows) and measured quantities (columns) in the AI measurement literature at the time of writing. Shaded cell = the cell this system occupies.

Unit / Quantity	Capability	Readiness/output	Influence
Model	LMArena Elo; model-capability evaluations	benchmark suites	(attention, informal)
Company	capability index (provider level)	compute analyses [Sevilla et al., 2022]	media lists
Country	Stanford AI Index [Maslej et al., 2024]; Global AI Index [Media, 2024]; Gov. AI Readiness [Insights, 2025]	—	—
Individual	—	h-index [Hirsch, 2005]; PageRank [Page et al., 1999]	h-index (proxy)
AI Originator	—	—	this system

Model level. Pairwise-comparison ratings such as LMArena’s Elo attach a score to a *model*, with an arbitrary scale anchor [Zheng et al., 2023]. The measured quantity is capability (or crowd preference); there is no notion of organizational or collective influence, and the scale is relative in the pairwise sense but not centered on a reference population in the aggregate sense used here. The Elo family is formally the closest relative (Appendix E compares the two scales); its level of analysis is the model, and models are not communities. A model that is a component of a larger Originator’s portfolio contributes to the Originator’s measurement only through the portfolio-level aggregation (Section 6.5); the unit is never the model itself.

Society / country level. Annual reports such as the Stanford AI Index [Maslej et al., 2024], the Global AI Index [Media, 2024], and the Government AI Readiness Index [Insights, 2025] measure

countries or societies on a yearly cadence. They are authoritative but slow, and they do not name the productive communities that create AI as measured units. Their influence on policy is real and desirable; their granularity is national, which is the wrong granularity for the question “which community is changing how society works.”

Company / list level. Company-level trackers (e.g., compute-based analyses of frontier training [Sevilla et al., 2022]) and media lists rank organizations or individuals; they are editorial or research artifacts rather than continuously maintained indexes. A media list may be rigorous, but it is not a measurement system: it has no stated composition rule, no conservation law, no settlement calendar, and no audit protocol. The difference between a ranked list and an index is exactly the difference between an editorial opinion and a repeatable measurement.

Individual-influence level. Citation-style measures such as the h-index [Hirsch, 2005] and PageRank [Page et al., 1999] quantify individual influence over an artifact graph (citations, web links). They are the closest formal ancestors of an influence index, and they inherit the well-documented cumulative-advantage dynamics of influence measures [Garfield, 1955, Merton, 1968]. None of them targets AI Originators as communities. Two lessons transfer directly into the design of FA: (i) influence measures exhibit path dependence (Matthew dynamics), so a measurement system must decide how much inertia is legitimate — here, the stock dimensions of FA carry inertia deliberately, while the live signal is centered so that discussion reorders but does not compound; and (ii) an influence measure that can be decomposed (the h-index decomposes into citation buckets) is far easier to audit than an opaque scalar.

2.2 The failure of AI benchmark evaluation as motivation

The strongest empirical argument for “capability $\neq$ influence” comes not from theory but from the recent record of benchmark evaluation failing at its own job. We collect the failure modes here because they are the concrete evidence that a capability score is a manipulable artifact of an evaluation pipeline, and therefore cannot serve as the public’s measure of where the field stands.

Contamination. Large public benchmark corpora are routinely exposed to models during pretraining or fine-tuning, so that “measuring” a model on a leaked benchmark measures memorization rather than capability. The documentation of the MMLU benchmark [Hendrycks et al., 2021] and of BIG-bench [Srivastava et al., 2023] discusses contamination risk explicitly, and a systematic study of closed-source LLMs found widespread leakage and evaluation malpractices across published leaderboards [Balloccu et al., 2024]. Koch et al. [Koch et al., 2021] document the reuse and recycling lifecycle of ML datasets, showing that a benchmark item born as a test item frequently re-enters the training corpus through downstream forks. The consequence is a measurement environment in which a high “capability” number can be manufactured without a real capability increase — precisely the Goodhart failure the system’s premise warns about.

Over-optimization and reward hacking. Even without leakage, optimizing directly against a proxy objective degrades the objective’s validity: this is the empirical content of reward over-optimization [Gao et al., 2023], reward hacking [Skalse et al., 2022], and spontaneous reward hacking in self-refinement [Pan et al., 2024]. The mechanism is generic — any score that is targeted stops being a measurement of the underlying quantity and becomes a bargaining chip between the optimizer and the scorer — and it applies with full force to public AI leaderboards, which are optimized against by every serious model developer.

Assessor bias. Even a clean benchmark requires a scoring function, and when the scoring function is another model (LLM-as-a-judge) it carries systematic biases: self-preference, position bias, and value contamination [Zheng et al., 2023, Wang et al., 2023, Santurkar et al., 2023]. These are not implementation bugs; they are the expected behavior of a model asked to evaluate its peers,

and they are the same class of value-contamination the system’s second argument (Section 1.2) predicts for any “which is better” assessment.

What the failure record implies for design. The record supports three design commitments that recur through this paper. (i) *Measure what is hard to fake*: an influence measure should lean on inputs whose manipulation cost is high — cumulative stock, cross-referenced third-party data, verified events — and treat fast, easily manufactured signals as a bounded, centered perturbation (Section 7.5). (ii) *De-average the sources*: because any single benchmark can be gamed, the FA capability block is a weighted mean over seven independent sources with an institutional replaceability mechanism (Sections 6.5 and 6.2); no single benchmark is decisive. (iii) *Separate the slow from the fast*: the contamination and over-optimization literature is largely a story about *fast* targets being gamed. The system’s two-component design — a slow, stock-like, multi-source FA anchor and a fast, centered, budgeted RI signal — is the structural response: it makes the gaming surface as small as the design can make it.

2.3 AI actor in governance literature

The governance literature uses the concept of an *AI actor* to allocate responsibility across the lifecycle of an AI system: deployer, provider, and developer roles appear in the EU AI Act [Union, 2024], the OECD AI Principles [Organisation for Economic Co-operation and Development, 2019], the NIST AI Risk Management Framework [of Standards and Technology, 2023], and the International AI Safety Report [Bengio et al., 2025]; measurement of AI systems is itself studied as a governance mechanism [Kwa et al., 2025, Shevlane et al., 2023], and agent-specific regulatory proposals extend the actor-centric trajectory to autonomous systems [Gardhouse et al., 2026, Hacker and Holweg, 2026]. In this literature “actor” is an instrument of responsibility assignment, not a measurement unit. The present system inherits the lifecycle framing but inverts its purpose: instead of assigning blame, it measures cumulative social influence. The two readings are complementary: governance actors are the responsibility-bearing instances of the Originator class, and a governance regime that needs to attribute responsibility to a developer also needs an account of that developer’s cumulative footprint. The system does not provide that account in governance form; it provides the measurement form, and the mapping between the two is left to the governance community.

2.4 Financial AI equity indexes as a formal precedent

Relative, centered, real-time indexing has a mature precedent in financial equity indexes, including industry-vertical equity indexes that track companies engaged in AI (for example, exchange- and index-provider-compiled AI industry indexes). These indexes establish that a *relative, center-anchored, real-time* scale can be operated publicly and be trusted for its measurement semantics. The present system borrows this form but changes the measured quantity: equity indexes measure market capitalization, i.e., the market’s pricing of a company; the Eins Index measures the social influence of an AI Originator. The novelty is therefore in the *combination* — measurement unit and measured quantity — not in the index form itself.

Three transferable lessons come from this precedent, and all three are adopted. *First*, a relative scale needs an explicit, reproducible composition rule and an explicit rebalancing calendar; the settlement calendar in Section 10.3 plays that role. *Second*, a center-anchored scale (an index “level” that does not drift for compositional reasons) requires a conservation or normalization discipline; the centering of RI in Section 5.6 is that discipline, and the no-backcasting rule of Section 3.5 keeps the level history auditable. *Third*, a real-time index needs a halt/suspend mechanism for integrity events; the system’s suspend–resume release protocol (Section 10.5) is that mechanism. The differences

are equally important and are stated as disclaimers (Section 13.6): equity indexes measure price discovery in a regulated market; the Eins Index measures a social-scientific construct with no market mechanism and no regulatory mandate, and nothing in the formal resemblance implies financial status.

2.5 Gap positioning and honest boundaries

Relative measures already exist at the model level (pairwise Elo). Therefore the precise claim is *not* “first relative index”; it is: the Eins Index is, to our knowledge from searchable public information, the first *relative, centered, real-time* measurement scale for the *social influence of AI Originators* as contributor-level units. Searches in both English and Chinese did not surface a comparable public-facing AI *influence* index at the time of writing. Two honest boundaries attach to this claim. First, “search has not found” is not “does not exist”; the claim is deliberately weak. Second, the system’s breadth is bounded by its sources: it measures influence *to the extent expressible in verifiable public data*, not influence in some unrestricted sense. “Comprehensive” is therefore not claimed; *measurable social influence bounded by verifiable public sources* is.

A third boundary is epistemic rather than empirical. The claim “first relative, centered, real-time influence scale for Originators” is a claim about the *combination*, and a combination claim is only as strong as its weakest component. The relative and real-time components have established precedent (Sections 2.4 and 7.3); the centered conservation discipline and the Originator unit are the components with the least precedent.

Part II

The Object

3 The Object: AI Originators and the Field of Measurement

3.1 Definition and class semantics

Definition 3.1 (AI Originator). An *AI Originator* is a community that continuously creates, releases, and iterates AI models, thereby acting as a source of intelligent capability that supplies AI productive capacity and technological innovation to human society. An AI Originator’s operational and technical boundary need not coincide with its capital, legal, or brand boundary.

The class definition is the load-bearing piece of the ontology. It is written to admit open-source collectives and joint initiatives: an Originator is identified by *what it produces and propagates*, not by its corporate form. This is what distinguishes the unit from “company” and “brand.” Three features of the definition are worth making explicit because they do the ontological work.

Continuity. “Continuously creates, releases, and iterates” is a diachronic predicate, not a synchronic one. A one-off release of a model does not make a community an Originator in the index’s sense; the predicate requires a sustained production stream. This excludes, by design, entities that are relevant to the field but are not *sources of model supply* — application vendors, consultancies, and aggregators that consume models without creating them. The definition thereby implements the boundary drawn in Section 3.6 between Originators (the supply side of AI productive capacity) and distribution channels.

Community, not person. “Community” deliberately includes (i) a corporate research lab, (ii) an open-source collective with no legal form, (iii) a joint initiative spanning several companies, and

(iv) in principle, a long-running research program within a university. The membership criteria of Section 3.3 then decide which of these are *measured*; the definition decides only which objects are *in principle* measurable.

Independence from legal form. “Operational and technical boundary need not coincide with capital, legal, or brand boundary” is what lets the roster track reality: a research lab that operates semi-independently of its parent, or a collective whose brand is held by a foundation, is a single Originator if its production stream is a single stream. Section 3.2 turns this into a resolution procedure.

3.2 Entity identity and attribution

Measurement requires that the same Originator be the same object across time and across sources, and that artifacts (models, repositories, posts, citations) be attributed to the correct Originator. This is a record-linkage problem in the sense of the entity-resolution literature [Christen, 2012], and it is not a philosophical footnote: every input to FA and RI is an attribution decision, and an attribution error propagates through the entire composition.

Primary-key discipline. Each Originator has a stable internal identifier, and every measurement input carries a reference to that identifier through a mapping layer. A model released by a partner laboratory is attributed to the Originator that funds and propagates it *if and only if* the production stream is the Originator’s; otherwise it is attributed to its own Originator or to an “unmapped” bucket. The mapping layer maintains the equivalent of a lookup table (name variants, aliases, historical names) and logs every mapping decision, so that attribution is a versioned, auditable artifact rather than a silent default. Mappings that cannot be resolved are kept in a separate, explicitly labeled bucket rather than folded into the nearest plausible Originator; this prevents contamination of the measured set by misattributed inputs.

Boundary of the identity procedure. The procedure is deliberately pragmatic — “good enough to scrutinize” — rather than legally definitive. It aims to make attribution (a) *consistent* (the same artifact maps to the same Originator across time), (b) *reversible* (a mapping error can be corrected and the correction propagated without rewriting history), and (c) *auditable* (every mapping has a source and a decision time). It does not aim to settle corporate-law questions about who “owns” a model, because the measured quantity is influence, not ownership.

Proposition 3.1 (Interior membership addition preserves existing normalizations). *Let $N(t)$ be the included set at time t , and let a new Originator u be admitted at $t' > t$. If for every factor under the linear rule (3) the new unit’s raw value does not exceed the current included maximum, then the normalized values — and hence FA — of all previously included units are unchanged by the admission.*

Proof. The linear rule (3) divides by the included maximum $\max_{j \in \mathcal{I}} v_j$. Adding u changes the maximum only if $v_u > \max_{j \in \mathcal{I}} v_j$, which the premise excludes; hence the denominator is unchanged and every previously included normalized value is unchanged. The result extends to any composition that is a deterministic function of the normalized values alone. $\square$

The proposition is the formal reason the system can grow without “restating” the past: an admission that does not redefine a normalization extreme is observationally invisible to existing members. When an admission *does* change an extreme, the system does not backcast (Section 3.5); it discloses the revaluation alongside the change. The boundary case where a new unit redefines an extreme is exactly the case where the index-world “no restatement on rebalancing” rule applies.

3.3 Membership and measurement eligibility

We separate two formal objects: the class (every community that releases models is an AI Originator by Definition 3.1) and the *membership set* of the index (which Originators are actually measured). Entering the index is governed by inclusion rules, not by the class definition. This separation lets the ontology be stable while the roster evolves under stated rules.

Three set-level objects are used throughout the formal framework (Section 6.1): the *universe* $\mathcal{U}$ (all AI Originators in principle measurable), the *included set* $\mathcal{I}(t)$ (the reference set that defines normalization extremes and the centering mean), and the *shadow set* $\mathcal{S}(t)$ (evaluated with the same parameters and displayed, but excluded from normalization extremes and the centering mean). The shadow tier is the index-world analogue of a watch list: it lets an Originator be evaluated and displayed before it qualifies for the reference set, without letting an unevaluated or partially covered unit distort the scale.

3.4 Inclusion rules

Admission to $\mathcal{I}$ is governed by stated rules, so that the roster is a consequence of the rules and the evidence rather than an editorial preference. The rules are versioned (the roster version appears in every published artifact; Section 10.2). The current rules are:

1. **Production-stream predicate.** The candidate must satisfy the continuity predicate of Definition 3.1: a documented history of creating, releasing, and iterating AI models, verifiable from public sources.
2. **Coverage predicate.** The candidate must have adequate measurement coverage in the sources the system relies on — formally, the fraction of FA factor fields with a valid observation must meet a stated threshold. This rule is what currently excludes organizations whose contribution is real but whose authoritative evaluation coverage is incomplete (Section 3.6); they are candidates for future admission as coverage grows.
3. **Identity predicate.** The candidate’s production stream must be resolvable to a single, stable identity under the procedure of Section 3.2; unresolvable attribution is grounds for exclusion or shadow status.
4. **Disclosure predicate.** The candidate must not present a legal or data-compliance obstacle to measurement (Section H.1).
5. **No-backcasting discipline.** Admission takes effect at a settlement; published history is not restated (Sections 3.5 and 10.4).

The inclusion rules themselves are subject to governance and revision (Section 10); changing a rule is a method change with a version vector entry, not a silent tweak. The rules are deliberately conservative in one dimension and deliberately permissive in another: conservative in that coverage and identity are hard gates, permissive in that legal form is not a gate (an open-source collective can qualify).

Admission is not closed to new entrants. The Arbitration Committee will open application registration in a later phase, and any AI Originator satisfying these inclusion rules is welcome to apply; the opening schedule is intended to be announced in due course.

3.5 Dynamic cross-section and no backcasting

The membership set is a dynamic cross-section: Originators may be added as they qualify, and each snapshot is computed over the membership set *at that time*. Constituent versions are stored. When rebalancing changes membership or when measurement methods change, the published series is *not* backcast: the historical series stands as it was published, and any enlargement-induced revaluation is disclosed alongside the change. This is the index-world analogue of the “no restatement on rebalancing” rule and keeps the published record auditable.

Two consequences of no-backcasting deserve emphasis. First, it makes the historical record a *chain of contemporaneous judgments*: a reader can in principle see what the system believed, when, and under which roster and method version. This is the property that supports external audit (Section 9.3). Second, it converts methodological honesty into a mechanical discipline: a method change cannot be smuggled into history as if it had always been true. The version vectors of Section 10.2 make the discipline checkable.

3.6 First assessment scope and defensive statements

The first assessment covers the set of AI Originators in the FA fundamental field: twelve communities, listed with their names in Appendix C. The roster is intentionally not asserted to be exhaustive. In particular, the system acknowledges the AI supply-chain contributions of organizations whose authoritative evaluation-source coverage is not yet sufficient for inclusion in the FA fundamental field; these are candidates for future inclusion under the inclusion rules. ByteDance is the current example at the organizational level. ByteDance is an AI Originator in the system’s universe: its Seed/Doubao model family has a documented release history verifiable from public sources, which satisfies the production-stream predicate of Section 3.4. It is not yet in the FA fundamental field because its measurable footprint is partial at the time of writing: its frontier models are served API-only without public weights; its public-weight releases are few and small, with a single mid-scale exception (Seed-OSS-36B, August 2025, Apache-2.0); and the authoritative evaluation sources the system relies on did not yet provide complete, stable coverage of its release stream. This is a coverage judgment under the inclusion rules, not a judgment of significance: ByteDance is a candidate for future admission as its coverage stabilizes, and exclusion from the FA field says nothing about the live-signal layer.

Application-layer AI productivity tools are not measured at any layer, for a definitional reason rather than a data reason. Coding assistants (Trae, Cursor, Copilot, Windsurf) and agentic IDEs (Codex) are distribution channels for influence, not source communities; an assessment method for application-layer influence is under study. Three properties carry the boundary. *First, origination.* A model is trained and released once by one community, and capability flows from that single release into many downstream products; the releasing community is therefore the attributable source — the AI Originator of Definition 3.1, the unit the FA field measures. Counting a coding assistant that routes several models would re-count the same underlying influence through each channel — a double-counting failure with a formal precedent in value-added accounting, which attributes each portion of value to its original source once [Koopman et al., 2014]. The system’s boundary defaults already encode this rule (originator-as-platform and model-as-application, Section 3.7): a thin wrapper over an Originator’s model does not create a new Originator. *Second, attribution tractability.* A model release is a public, versioned, comparable artifact with a stable identity under the procedure of Section 3.2; an application is a moving, partly closed target whose model routing changes without a stable attributable artifact, so per-product evaluation would have to be repeated at every re-route. The boundary is drawn on the model layer because that is where influence is

attributable, not because applications are irrelevant to influence. *Third, scope symmetry.* Tools built by measured Originators follow the same rule: **Codex** is built by OpenAI and **Claude Code** by Anthropic, both measured in the FA field; the tool adds distribution, not a new Originator, and the influence it carries is already attributed to the model’s community. The converse case — a tool whose maker is not itself in the FA field, such as **Trae** by ByteDance — is covered by the same rule: the maker’s organizational eligibility is assessed separately, while the tool itself remains a channel.

The defensive statements are precise about *which field* they concern. The roster is the FA *fundamental field* — the set of Originators measured on the slow anchor. The RI field is broader by construction: the live signal is measured over any Originator that appears in the verified public information the system consumes (Section 5.4), whether or not it is in the FA roster. An organization can therefore appear in the live-signal layer without being in the FA fundamental field. The defensive statements concern only the FA field.

3.7 Boundary cases

Boundary cases — an originator that is also a platform, a model that is also an application, a collective with no legal form — are handled by rules of thumb that are “good enough to scrutinize” rather than by legal-grade adjudication. The criteria are not legally binding; they are written to be examined, objected to, and improved. Three boundary cases recur in practice and have a stated default.

Originator-as-platform. When an Originator also operates a platform that hosts other models (a hosting, aggregation, or serving layer), only the Originator’s own production stream counts toward its measurement; the hosted artifacts of others are attributed to their own Originators. This default prevents double counting: the platform value of hosting is an ecosystem service, not a model-production stream.

Model-as-application. A model that ships inside an application is measured as a model of its Originator; the application’s distribution value belongs to the channel, which the FA field does not measure (Section 3.6). The boundary is drawn at the model layer: if a product is a thin wrapper over an Originator’s model, it does not create a new Originator.

Collective-without-legal-form. A collective is admitted if its production stream, identity, and coverage satisfy the inclusion rules; the absence of a legal person is not an obstacle (Section 3.4). The harder question — when a collective’s production stream merges with or splits from another’s — is resolved by the identity procedure of Section 3.2: the rule is “one measurable production stream per Originator,” and the evidence is the verifiable record of releases and attribution.

Part III

The Measure

4 Measurement Theory and Construct Validity

4.1 The measurement-theoretic stance

“Influence on society” is not a property that can be read off the world with an instrument, the way length is read off a ruler. It is a *theoretical construct*: an inferred attribute whose reality is evidenced by the pattern of observables it explains [Cronbach and Meehl, 1955, Messick, 1995]. This section takes the construct-validity tradition seriously, because an influence index that ignores it is either (a) an unexamined bundle of arbitrary weights or (b) an overclaim that its number is “the” truth. The

system’s answer is: the number is a measurement of a construct, the construct is explicitly defined, the operationalization is derived from the definition rather than from convenience, and the evidence that the operationalization tracks the construct is stated as a research program with acceptance criteria (Section 8.6).

The stance has three consequences for the measure. *First*, validity is a property of *inferences*, not of the number itself [Messick, 1995]: the same scalar can be a valid measure of one construct and an invalid measure of another. The paper therefore always asks “valid for what inference?” — and the primary inference is *cross-sectional comparison of Originators’ social influence within a defined reference set*. *Second*, validity is a matter of evidence accumulated over time, not a one-time declaration. The acceptance criteria in Section 8.6 are therefore framed as a running program with explicit evidence gates. *Third*, the construct is multidimensional, so the measurement system is a composite; composite construction has its own methodology and failure modes [Nardo et al., 2005, Saisana et al., 2005], which Section 8.7 treats as a quantitative research design.

4.2 Construct validity

Cronbach and Meehl [Cronbach and Meehl, 1955] introduced *construct validity* for the case where no single criterion exists against which a test can be validated: one validates a construct by testing a network of theoretical predictions about observables. Campbell and Fiske [Campbell and Fiske, 1959] added the operational discipline of the multitrait–multimethod matrix: a measure should converge with other measures of the *same* trait (convergent validity) and diverge from measures of *different* traits taken by the *same* method (discriminant validity). Messick [Messick, 1995] extended the notion to include the consequences and values of measurement: a valid measure is one whose score meaning and use are defensible, which forces the score’s value assumptions into the open rather than hiding them.

The system maps these three layers onto its own objects as follows.

Nomothetic network. The construct “social influence of an AI Originator” enters a network of predictions: it should (i) be positively but imperfectly correlated with capability (necessary-not-sufficient, Section 1.2); (ii) track observable consequences of use and adoption (ecosystem stock, commercial reach, discussion volume); (iii) be relatively stable over short horizons (the anchor) while responsive over longer horizons (the settlement calendar); and (iv) be decomposable into the four channels of Section 5.1 in a way that explains the cross-sectional pattern. Each prediction is a testable claim about the correlation structure of the observables, and the acceptance criteria of Section 8.6 are the tests.

Multitrait–multimethod discipline. The system operationalizes each channel through *multiple, methodologically distinct sources* (Section 6.5): capability through several independent benchmarks, ecosystem through repository and citation and download sources, commerce through price and adoption sources. This is the composite-indicator analogue of the multimethod matrix: the score is trustworthy to the extent that methodologically distinct measures of the same channel agree (convergence), while measures of different channels do not collapse into one another (discriminability). A regression of the flagship on its own channel scores cannot establish this — the channels are the composition’s inputs by construction — so the convergent and discriminant evidence must come from *external* measures that are not inputs: e.g., independent surveys, downstream adoption statistics, or events. The research program of Section 8.6 is designed around external evidence for exactly this reason.

Consequential validity. The system does not claim value-neutrality; it claims *transparency of values*. Section 5.2 exposes every weight as a commitment about what influence is, and Section 8.9 states the boundaries of the score’s meaning. The consequential question — what a real-time public

influence number does to the field it measures — is treated as an open empirical question and as a design constraint at the same time: the centering of the live signal and the conservation law (Section 5.6) are consequential-validity mechanisms that limit the score’s capacity to manufacture its own narrative (Section 7.5).

Proposition 4.1 (Construct–indicator consistency of the influence definition). *Let the construct “social influence” be defined as the four-channel composite $\langle \text{use}, \text{discussion}, \text{iteration}, \text{diffusion} \rangle$ (Section 5.1). Then every block of the FA composition (Section 6.5) and every track of the RI signal (Section 6.6) is assigned to exactly one channel, and each channel has at least one block or track assigned to it.*

Proof. By construction of Tables 3 and 4. The capability and edge blocks realize the necessary-but-insufficient capability input to *use*; the ecosystem block realizes *iteration* and the stock side of *use*; the commerce block realizes the accessibility side of *use* and *diffusion*; the RI signal realizes *discussion*; the absolute-heat line realizes the magnitude of *discussion* independently of polarity. $\square$

The proposition is the paper’s formal answer to the question “where do the weights come from?” The *structure* of the composition comes from the definition (content validity, Section 4.4); the *numerical weights* are V1 production values calibrated by the expert group and engineering tuning and frozen at release (Appendix B), with a unified calibration experiment listed as future work (Section 12.2). The structure is a claim about meaning; the weights are an empirical calibration. Conflating the two is the classic composite-indicator error [Leamer, 1983, Nardo et al., 2005].

4.3 The construct: influence on society

Before operationalization, the construct itself needs boundaries. “Influence on society” is restricted in three ways that prevent the measure from silently meaning something else.

Restriction to the measured channels. Influence, as measured, is the composite of use, discussion, iteration, and diffusion as they are *observable in verifiable public sources*. This excludes, by declaration: influence through private lobbying, influence through regulatory capture, and influence through closed customer relationships that leave no public trace. These are real channels of influence that the system does not claim to measure; they are boundaries, stated as such.

Restriction to social, not economic, impact. The system measures a social-scientific construct, not a financial one. The commerce block is included because *accessibility* (can people reach and afford the technology?) is a channel of social diffusion, not because the system prices companies. The disclaimer (Section 13.6) is not boilerplate; it is the boundary enforcement of the construct definition.

Restriction to current-period observable consequences. The system measures influence that has *already happened and left a trace*, not predicted influence. Forward-looking influence (the potential of a model that has not yet diffused) is not measured; the index reflects what has already occurred (Section 1.6).

4.4 Content validity

Content validity asks whether the operationalization *covers the construct* rather than a convenient subset of it. The argument has two parts.

The definition-to-block mapping is exhaustive. Proposition 4.1 and Table 3 show that every channel of the definition is represented. The converse — that no block represents a channel outside the definition — is enforced by the “composition–vision correspondence” review of Section 5.2: each block

has a stated channel, and a block without a channel is a design error. The one historically contested case, a media/attention dimension inside the anchor, was removed in V1 because its semantics overlapped the discussion channel carried by the live signal; removing it is itself a content-validity decision (two channels must not be counted twice).

The within-channel coverage is reasonable. Capability is measured by seven independent benchmarks spanning general knowledge, coding, and crowd preference (Appendix B); ecosystem by repository, download, citation, and adjudicated-ledger sources; commerce by value, price, free access, and adoption momentum. The coverage is deliberately wider than any single source would give, and Section 8.7 quantifies how sensitive the composite is to individual factors. Content validity does not require perfect coverage; it requires that the coverage be a defensible sample of the construct’s domain, and that coverage gaps be disclosed (Section 11.3).

4.5 Convergent and discriminant validity

The multitrait–multimethod logic of Campbell and Fiske [Campbell and Fiske, 1959] translates into a set of *predictions* that an external observer can check:

1. *Convergent: method change, same channel.* Two different measurement methods for the same channel (e.g., two independent coding benchmarks within the capability block) should agree in their cross-sectional ordering more strongly than two methods for different channels.
2. *Convergent: channel to external consequence.* Channel scores should correlate with *external* consequences that are not inputs to the system — for example, independent adoption statistics or third-party usage surveys.
3. *Discriminant: capability vs. ecosystem.* The capability and ecosystem channel scores should *not* be near-perfectly correlated across the roster; if they were, the composition would be a disguised single factor and the weights would be meaningless. The first snapshot already shows the discriminant structure (Section 11.2): capability and ecosystem rankings diverge sharply.
4. *Discriminant: anchor vs. live.* The FA anchor and the RI signal should move on different time scales; the anchor should be comparatively stable within a settlement window while the live signal moves. This prediction is operationalized as the time-scale separation test of Section 8.6.

Table 2 summarizes the planned external evidence sources. The distinctive feature of the table is that none of the evidence is generated by the system itself: the convergent evidence comes from sources that the composition does not consume, and the discriminant evidence comes from patterns the composition cannot impose.

4.6 Criterion and known-group validity

Where a criterion exists, construct validity is supplemented by criterion validity: the score should relate to a criterion in the predicted way. Two criterion designs are feasible and are planned.

Known-groups design. The system should order Originators in a way that matches an independently defensible ordering of “social footprint.” A natural criterion group is the set of Originators with large verified user bases and ecosystem footprints; the score should place them in the top tier without being told to. This is weak evidence, but it is evidence: a score that ranked a no-evidence organization above a documented-ubiquitous one would be falsified regardless of its internal algebra.

Event-consequence design. A verified disruptive event (a widely adopted release, a substantive incident) should move the relevant sub-components in the predicted direction and magnitude:

Table 2: Construct-validity evidence plan: external (non-input) evidence for the convergent and discriminant structure of the score. “External” means the evidence is not an input to FA or RI.

Prediction	External evidence class	Status
Convergent (channel-level)	independent coding/agentive evaluation suites not in the capability block; third-party usage surveys	planned
Convergent (aggregate)	independent organizational trackers; expert panels (Section 8.6)	planned
Discriminant (channels)	divergence of capability vs. ecosystem vs. commerce rankings across the roster	§11.2, preliminary
Discriminant (time scale)	within-window stability of FA vs. movement of RI	planned (needs longer window)

capability and ecosystem stock on release, the live signal immediately, and the anchor only through the settlement adjustments (D , R). The event record provides the criterion. Because no qualifying disruptive event has yet been observed within the operating history at writing, this design is planned rather than evidenced.

4.7 Threats to validity and design responses

The standard validity threats have names, and the system has a response to each; the response is what separates a defensible measurement from an optimistic one.

Construct underrepresentation. The operationalization may capture only part of the construct. Response: the four-channel definition is the declared coverage boundary, and the restriction to verifiable public sources is stated (Section 4.3); underrepresentation is disclosed rather than hidden.

Construct-irrelevant variance. The score may be moved by things that are not the construct: benchmark contamination [Balloccu et al., 2024], assessor bias [Zheng et al., 2023], attribution noise, or source-specific artifacts. Response: the multisource, multi-method design (Sections 6.5 and 6.2), the missing-value taxonomy (Section 6.4), and the audit inequality (Section 6.9) are the operational responses; each places a hard bound on how much irrelevant variance a single artifact can inject.

Goodhart drift. Once the score is a target, the measured quantity changes (Sections 1.2 and 7.5). Response: this is the deepest threat and the one the architecture addresses most structurally — slow multi-source anchor, centered bounded live signal, asymmetric disclosure. The system does not claim to be immune; it claims that the marginal cost of moving the score is engineered to be high.

Value contamination. The weights embed values. Response: Section 5.2 exposes every weight as a commitment; the acceptance framework treats the exposure itself as evidence (Section 8.3).

Reactive feedback. A public score changes the behavior of the measured. The centering and the no-backcasting discipline (Section 3.5) are the structural responses; whether they are sufficient is itself an empirical question, and the reflexive dynamics are discussed in Section 7.5.

5 From Influence Definition to Composition

5.1 From definition to decomposition

Section 1.2 defined influence as the composite $use \times discussion \times iteration \times diffusion$. This section turns that definition into an operational decomposition: a slow *Fundamental Anchor* (FA) component that carries the cumulative, hard-to-manipulate part of influence, and a fast *Real-time Influence* (RI) component that carries the discussion-and-diffusion part. Table 3 is the load-bearing map: it assigns every block and every track to a channel of the definition. Each formula is an implementation of the map, not an arbitrary aggregation.

Table 3: Composition–vision map: every FA block and every RI track assigned to the channel of the influence definition it operationalizes. “Necessary-not- sufficient” marks the capability input.

Component	Channel	Structural role	Evidence for design
Capability block	use (necessary, not sufficient)	seven independent benchmarks; block weight deliberately bounded	§5.2
Edge block	use $\times$ diffusion (accessibility)	small-model coverage in four parameter bands	§5.3
Ecosystem block	iteration $\times$ use (stock)	developer stock, downloads, citations, adjudicated ledger; no time decay	§5.3
Commerce block	use $\times$ diffusion (reachability)	value, price, free access, adoption momentum	§5.2
Sustained leadership D	use $\times$ iteration (persistence)	time-verified frontier hold, capped	§5.3
Real-harm correction R	(harm channel, negatively)	verified substantive incidents only	§7.2
RI signal (signed)	discussion (reordering)	centered renewal envelope	§6.6
Absolute heat H	discussion (magnitude)	same envelope, absolute values	§6.7

The composition weights are *not* arbitrary: each component is anchored in the vision-derived definition above. The numerical values are V1 production values, calibrated by the expert group and engineering tuning and frozen at release (Appendix B); a unified calibration experiment is listed as future work (Section 12.2). Two things are deliberately separated: the *structure* of the composition (a claim about meaning, defended by Proposition 4.1) and the *numerical weights* (an empirical calibration, to be stress-tested in Section 8.7).

5.2 Composition and its correspondence to the vision

The FA composite is a weighted sum of four blocks plus two adjustments. The semantic anchor of each weight is as follows:

Capability (capability block). Capability is *necessary but not sufficient* for influence. Its block weight is deliberately bounded: a perfect capability score determines only part of the anchor — the ecosystem, edge, and commercial blocks together retain enough weight to reshape the ranking — and within the capability block a seven-source weighted mean keeps no single benchmark decisive. This is the structural evidence of “capability $\neq$ influence.”

Edge coverage (edge block). Small, capable models run on edge devices and in local deployments, putting AI in front of people who would otherwise not use it. This block carries the *accessibility* / *diffusion* side of influence.

Ecosystem stock (ecosystem block). Use and iteration leave measurable stock: open-source repositories, downloads, models, citations, and an adjudicated ledger of research artifacts. This is the *use* $\times$ *iteration* side, accumulated over time with no time decay (Section 5.3).

Commercial accessibility (commerce block). Whether models are affordable and reachable as services shapes how widely they are used: value (capability per unit price), price, free access, and adoption momentum.

Sustained leadership (D). A time-verified lead — holding frontier ranks across consecutive settled periods — is a slow adjustment to FA, capped at a fraction of FA (Appendix B). It measures *sustained* control of the frontier rather than a snapshot. Falling out of all thresholds sets D to zero immediately, with no decay tail: the inertia that would otherwise be double-counted is already carried by the stock dimensions (Property 6.20).

Real-time influence (RI). Discussion, attention, and diffusion are the fast channels of influence. The RI line is *centered* (Section 5.6): it reorders positions within a conserved total but cannot, by itself, enlarge the pool.

Real-harm correction (R). Verified, substantive incidents — official data breaches, critical security vulnerabilities, regulatory penalties — are the only channel through which harm lowers the anchor, and only when officially confirmed, with sources attached and a correction channel open (Sections 5.3 and 7.2).

Removing the media/attention dimension from FA and centering RI are two sides of one boundary claim: *discussion volume cannot, by itself, prove influence*. Attention is a channel of influence, not its certification. The structural reading of this claim is worth stating explicitly: because attention is a channel, it must enter the measurement somewhere — it enters as the RI signal, centered and bounded, so that it reorders but cannot certify. And because attention is not a certification, it must not enter the anchor as a slow stock — hence the removal of the media dimension from the V1 FA composition. The two moves are one policy implemented in two places.

5.3 FA mechanism: the slow anchor

The FA value of an Originator i is computed as a weighted mean over its present dimensions, on a unified scale, using the production composition of Section 6. Three properties make FA a “long-run fairness skeleton”:

1. **Stock, not flow, with no time decay.** Every FA input is a monotonic stock quantity (evaluation score, count, cumulative citation, ledger). Time decay lives *only* in RI; FA does not forget past contributions on a timer. This is deliberate: an anchor should move slowly. The counterargument — that “newer” contributions should weigh more — is answered structurally: the *iterative* novelty of contributions is carried by the adjudicated ledger’s tiering and by the live signal, not by decaying the anchor. A decaying anchor would erase the very cumulative influence the system exists to measure.
2. **Continuous-missing-value protocol.** Missing values are classified (Section 6.4) as true-zero, non-punitive absence, or leaderboard floor; a missing source never zeroes an Originator. Data-continuity governance carries the most recent valid value forward with an alert; an unavailable source does not silently produce a zero.

3. **Settlement and adjustment.** FA is settled twice daily at closed-market windows (UTC 04:00–07:00 and 16:00–19:00), on the same structure and with equal weights for both windows. The adjustments D and R apply only at settlement. During market hours, the settled FA value is frozen and consumed by the index unchanged.

5.4 RI mechanism: methodological description

The RI signal measures discussion and diffusion in near real time. The full algebraic structure is given in Section 6.6; here we state the design principles, without numerical parameters.

- **Source classes.** RI consumes independently verified authoritative media and public-platform information, aggregated to the Originator level. Individual source identities are deliberately not disclosed (Sections 13.3 and 13.6).
- **Heat and direction are separated.** A search index or a trending rank proves that *attention rose*; it cannot by itself certify whether the attention is favorable or unfavorable. Direction is assigned by a polarity model with an attention baseline, and only negatively *factual* signals count as negative.
- **Leaderboard first, topic backfill.** Ranking positions are the primary signal; discussion topics fill in where no rank is available. A non-ranking *attention* feed is deliberately excluded.
- **Half-life tiers.** Events are classified into content-type tiers (rank-driven, factual, opinion, emotional), each with its own half-life governing decay. Values are not disclosed.
- **No-stimulus decay to zero.** With no new stimulus, a pulse decays monotonically to zero and is then removed from the active set; a decayed pulse does not deposit into FA.
- **Renewal (contract rollover).** When a new event carries a full value at least as large as the residual of the old pulse, the old pulse is voided and the new value starts a fresh pulse; when it is smaller, the old pulse simply decays. Instantaneous value thus depends only on the current state, not on the arrival path (Axiom 6.6).
- **Eight-component decomposition.** The signed RI signal is decomposed into eight tracks — for each of the four content tiers, a positive and a negative component (Table 4). The decomposition is an identity (the sum of the tracks equals the signed signal at every time) and it is what makes the dual-track audit of Section 9.3 possible.
- **Budgetary constraints.** A volatility budget caps the total RI magnitude; a per-Originator per-settlement movement bound triggers an audit alert without truncating the value.

Table 4: The eight-component RI track decomposition (structural; no parameter values disclosed). Rows are content tiers, columns are signs; each track is a decaying-pulse sum over its own active events.

Tier	Positive track	Negative track	Decay kernel	Note
Rank-driven (board)	$RI_{+,1}$	$RI_{-,1}$	$2^{-(t-t_e)/T_{1/2}(1)}$	platform ranks, heat-weighted
Factual	$RI_{+,2}$	$RI_{-,2}$	$2^{-(t-t_e)/T_{1/2}(2)}$	verified events, longer tail
Opinion	$RI_{+,3}$	$RI_{-,3}$	$2^{-(t-t_e)/T_{1/2}(3)}$	authoritative assessments
Emotional	$RI_{+,4}$	$RI_{-,4}$	$2^{-(t-t_e)/T_{1/2}(4)}$	fast rise, fast decay

5.5 The four elements of a measurement scale

A measurement scale must provide four things: a *common scale* across units; a *stable center* that anchors comparison; *domain-universal* applicability (the same operations work for every unit); and *comparability* across time. The design maps these onto three properties: *relative* (common scale, by cross-sectional normalization), *centered* (stable center, by the conservation law), and *real-time* (comparability across time, by the collection–pricing–display pipeline). The formal definitions are in Section 6; the trust arguments are in Section 7. The four-element framing matters because “relative” alone is too weak a claim: an Elo rating is relative yet its scale anchor is arbitrary and it has no conservation structure. The distinctive claim is the *combination* — a common scale, a stable center, and a real-time cadence operating together, so that the number has the semantics of a measurement rather than the semantics of a score in a game.

5.6 Conservation and centralization: statement

The flagship index Θ is a centered composition:

$$\Theta_i(t) = \text{FA}_i^{\text{adj}}(t_0) + k_{\text{ri}} (\text{RI}_i(t) - \overline{\text{RI}}(t)) + \text{BR}_i(t), \quad (1)$$

where $\overline{\text{RI}}(t)$ is the cross-sectional mean of RI over the included set $\mathcal{I}(t)$, t_0 is the last closed-market settlement time, and BR is a reserved bridge slot that is zero except during a member-set transition (Section 6.10). The live-signal dispersion coefficient k_{ri} is reserved. Two consequences follow at once and are proved in Section 6.9:

- *Exact conservation on the adjusted anchor.* The sum of Θ over the included set equals the sum of FA^{adj} over that set, at every time (Proposition 6.9).
- *Approximate conservation on base FA.* Because the published index carries the sustained-leadership adjustment, the sum of base FA is only *approximately* equal to the sum of Θ , with a deviation bounded by the adjustment caps; it is exact when no Originator holds a leadership streak and no substantive incident is recorded (Proposition 6.11).

The design history of (1) is itself a piece of evidence. An earlier formulation added the RI term *uncentered*; in live validation it was discovered that this allowed the total pool to drift — transient discussion could inflate the aggregate, violating the “heat reorders but does not enlarge” principle — and the formulation was replaced at a settlement with the centered form. The centering is therefore not decoration: it is the mechanism that enforces the boundary claim of Section 5.2 at the level of the total pool.

The index family also includes an absolute-ability family (Γ, Λ) and sector aggregates (μ_Γ, M_Γ) that preserve cross-sectional ranking (Section 6.7); these exist so that relative measurement and absolute progress are not conflated (Proposition 6.3).

5.7 Counterarguments and design trade-offs

Every major design choice in the measure has a serious counterargument, and each is answered; the choices are deliberate, not oversights.

“*The weights are arbitrary.*” Answered at two levels. Structurally, Proposition 4.1 ties every block to the influence definition, and Table 3 makes the tie explicit. Numerically, the weights are V1 production values subject to a planned unified calibration experiment (Section 12.2) and to the sensitivity analysis of Section 8.7, which will report how rankings move under weight perturbation. Arbitrariness would mean “no reason”; the paper’s claim is “a stated reason and a planned test.”

“*The anchor is too slow to react to disruptive releases.*” The anchor is supposed to be slow; the live signal is the fast channel. A disruptive release moves the live signal immediately and the anchor at the next settlement through the capability re-evaluation. The two-layer design is the answer to the speed objection, not its victim.

“*A centered live signal punishes quiet excellence.*” The centering makes the live signal a zero-sum reordering, so an Originator with stable, high attention is at the mean and receives no bonus or penalty from the live term. That is intended: the live term is a *deviation* channel, and persistent influence is carried by the anchor. Punishing quiet excellence would require the live term to be absolute, which would reintroduce the pool-inflation failure the centering fixes.

“*Two settlement windows a day is arbitrary.*” The windows (UTC 04:00–07:00 and 16:00–19:00) are chosen to bracket the two hemispheres’ main activity periods and to bound how stale the frozen anchor can get within a day. The count and placement are engineering choices, revisable under governance (Section 10.3); what is not revisable without a method change is the discipline that the anchor freezes between settlements.

Part IV

The Mathematical Shell

6 Formal Framework: The Mathematical Shell

This section gives the production formulas in symbolic form. Numerical values of all FA-related parameters are in Appendix B; RI parameters are symbolic throughout and are not disclosed. The product codes are labels bound to symbols in Table 11 (Appendix A); formulas are stated entirely in symbols.

6.1 Indicator space and the layered observation state

Let $\mathcal{U}$ be the universe of AI Originators, $\mathcal{I}(t) \subseteq \mathcal{U}$ the *included* set at time t (the reference set that defines normalization extremes and the centering mean), and $\mathcal{S}(t) \subseteq \mathcal{U} \setminus \mathcal{I}(t)$ the *shadow* set (evaluated with the same parameters, displayed but excluded from normalization extremes and from the centering mean).

The system is a layered observation state. Each layer is a distinct kind of object with its own versioning, and the pipeline (Section 9.2) is a deterministic map between layers:

- L1 — raw observations.** For each Originator i and source s , a raw observation $x_{i,s}(t)$ with an observation time, a quality state, and a source version. L1 is the only layer that touches external reality directly; it is never retroactively edited, only corrected by new observations.
- L2 — derived indicators.** A registry of derived series $y_k(t)$ computed from L1 by stated formulas under the unified scale contract (Section 6.3) and the missing-value taxonomy (Section 6.4). Examples: the seven capability inputs, the ecosystem sources, the commerce factors, the settled FA blocks, and the RI track decomposition (Table 4).
- L3 — published indexes.** The index family (Section 6.7): a small set of public artifacts computed from L2 by the composition formulas. This is the layer the presentation layer (Section 6.8) projects onto public artifacts.

Definition 6.1 (Full indicator state). For each Originator i , let $x_i = (x_{i,1}, \dots, x_{i,K})$ be its vector of *L1 source inputs* — raw measurements from the third-party and self-collected sources of Section 13. The *full indicator state* at time t is the tuple of all raw inputs, normalized values, block composites, settled anchors, live signals, and derived index values for all $i \in \mathcal{U}$.

The layering is not cosmetic. It is what makes the system *recomputable in principle*: the published L2 formulas allow the L3 indexes to be checked against the L1 observations, without access to proprietary code. The layers also localize corruption and error (Section 6.2): a source outage at L1 degrades only the L2 indicators that consume it, and the audit protocol (Section 9.3) verifies each layer’s invariants separately.

6.2 Source mapping and circuit breakers

Every L2 indicator consumes a declared set of L1 sources, with weights and a replacement policy, so that the system’s dependence on any single source is an explicit, auditable graph rather than an implicit default. Two mechanisms make the dependence safe.

Independent circuit breakers. Each source has an independent quality gate (freshness, completeness, availability, schema validity, and jump checks). When a gate trips, the affected source’s contribution is degraded or suspended — a missing or degraded source never zeroes a factor and never silently fills with a suspected bad value; the most recent valid observation continues with an alert (Section 9.1). Because breakers are per-source, an outage is isolated to the smallest fault domain possible, and the rest of the measurement continues.

Weighted multi-source redundancy. Where several sources measure the same quantity (e.g., the seven capability benchmarks), the factor is a weighted mean, so no single source is decisive. This is the same redundancy logic as the composite indicator literature recommends against indicator-level sensitivity to any single input [Nardo et al., 2005, Saisana et al., 2005]. The combination of circuit breakers and redundancy gives a quantitative guarantee.

Proposition 6.1 (Circuit-breaker dilution bound). *Let a factor $\hat{v}$ be a weighted mean $\sum_s w_s y_s$ over sources s , with $\sum_s w_s = 1$ and $y_s \in [0, 1]$. If source s_0 is degraded (its observation dropped and the factor re-normalized over the remaining present sources), then the change in $\hat{v}$ satisfies*

$$|\hat{v} - \hat{v}_{\text{out}}| \leq \frac{w_{s_0}}{1 - w_{s_0}}, \quad (2)$$

where $\hat{v}_{\text{out}}$ is the factor recomputed over the surviving sources.

Proof. When s_0 is dropped, the surviving weights are re-normalized to sum to one, so $\hat{v}_{\text{out}} = \sum_{s \neq s_0} \frac{w_s}{1 - w_{s_0}} y_s$. The difference is $\hat{v} - \hat{v}_{\text{out}} = w_{s_0} (y_{s_0} - \sum_{s \neq s_0} \frac{w_s}{1 - w_{s_0}} y_s)$, and the bracket lies in $[-1, 1]$ because it is the difference of two values in $[0, 1]$. The bound follows. $\square$

The bound is the formal content of “a single-source outage dilutes rather than zeroes a factor”: with the largest capability-source weight (0.40, Appendix B), dropping it changes that factor by at most $0.40/0.60 \approx 0.67$ in the worst case, and in practice far less because the surviving sources remain informative. The mechanism is the institutional guarantee that the system is *not in competition with its sources* and that no single data provider holds a veto (Section 7.2).

6.3 The unified scale contract and robust normalization

All factor values are normalized by one of four rules, with extremes defined *only* over the included set $\mathcal{I}$ (shadow units use the same extremes):

$$\text{linear: } \hat{v} = \frac{v}{\max_{j \in \mathcal{I}} v_j}, \quad (3)$$

$$\text{heavy-tail: } \hat{v} = \frac{\log_{10}(1+v)}{\log_{10}(1+\max_{j \in \mathcal{I}} v_j)}, \quad (4)$$

$$\text{inverse: } v \mapsto \frac{1}{v} \text{ then (3) or (4),} \quad (5)$$

$$\text{boolean: } \hat{v} = v \in \{0, 1\}. \quad (6)$$

Every factor is normalized *before* any weighting (the “unified scale first” discipline). This is a deliberate departure from composite-indicator practice that normalizes at the end of aggregation: because every sub-factor is already in $[0, 1]$ with the same semantics (fraction of the included-set maximum), weighted sums of sub-factors are comparisons of like quantities, and pathological behaviors such as negative composites or log-domain failures are impossible by construction.

Robustness. Raw source values are often heavy-tailed and noisy. The design uses three robustness layers. *First*, the heavy-tail rule (4) is itself a compression that de-emphasizes extreme right tails — the appropriate treatment for skewed counts such as downloads and citations. *Second*, anomaly detection uses a robust scale (the median absolute deviation, scaled to estimate the standard deviation) rather than the raw standard deviation, following the recommendations of the robust-statistics literature [Rousseeuw and Croux, 1993, Huber and Ronchetti, 2009]; a single contaminated observation inflates the MAD far less than it inflates the variance, so anomaly flags are not triggered by outliers themselves. *Third*, verified real jumps are never winsorized or smoothed: the robustness layers protect the *scale* from contamination, they do not suppress true dynamics (the data-continuity protocol of Section 9.1 makes this distinction explicit).

Proposition 6.2 (Robust scale is bounded under a single outlier). *Let z be a sample of size n and let MAD be its median absolute deviation from the median. Replacing one observation by an arbitrarily large value changes the robust scale 1.4826 MAD by a bounded amount, whereas the sample standard deviation can increase without bound.*

Proof. The median of z is the $\lceil n/2 \rceil$ -th order statistic; moving a single observation to $+\infty$ can shift the median by at most one position in the order statistics. The absolute deviations from the median therefore change by a bounded amount on a bounded number of points, so their median — and hence the MAD — is bounded. The standard deviation, by contrast, grows like the moved value divided by $\sqrt{n}$, which is unbounded. $\square$

The proposition formalizes why the anomaly pipeline (Section 9.3) can flag contamination without being blinded by it: the scale that defines “anomalous” is robust to the very outlier being flagged.

Proposition 6.3 (Ratio invariance of linear factors). *For any factor under the linear normalization (3), a common proportional change $v_j \mapsto cv_j$ for all j , with $c > 0$, leaves every normalized value and hence FA unchanged.*

Proof. $\widehat{cv_j} = cv_j / \max_k(cv_k) = v_j / \max_k v_k$. $\square$

Remark 6.1. *The heavy-tail rule (4) contains an additive 1 inside the logarithm, so it is only approximately scale-invariant. Absolute progress is not suppressed by this design: it is reported through the absolute family (20)–(23), which is exactly the separation the paper argues for.*

6.4 Missing-value taxonomy and two-level renormalization

Missingness is the most common corruption vector in real measurement, and the system treats it with the formality it deserves, following the missing-data taxonomy of Rubin [Rubin, 1976, Little and Rubin, 2019]. The design distinguishes four missingness states for the FA field, which refine the earlier trichotomy into a four-state taxonomy whose judgement is registered and audited:

True zero (*P*-state). The object of measurement is *participation itself*, and the unit is present but has no observable activity of the measured kind (e.g., a present Originator with no official free API access, no open-source model, no edge-band coverage). Handling: the zero is *counted with weight zero and always present*; it is forbidden to re-normalize it away. A true zero is a fact about participation, and erasing it would reward absence.

Numerical absence (*Z*-state). The source does not cover the unit, or the value is unknown/unbounded (e.g., the source’s editorial list does not include the unit; a data field is frozen; no routing exists). Handling: *non-punitive renormalization over the present dimensions*. This is the Rubin-compatible reading: if missingness is independent of the unobserved value given the observed, dropping the dimension and re-normalizing is unbiased.

Leaderboard floor (*A*-state). The unit is below the printed cutoff of a ranked source; “it is not that it is absent, it is that it was not photographed onto the board.” Handling: *value by the last readable rank*, taken dynamically from the current board tail — never a hardcoded constant. This prevents the floor from becoming a privileged “absence discount.”

Technical unavailability (*F*-state). The observation is missing for an infrastructural reason (source outage, schema failure, transmission error). Handling: the most recent valid value is carried forward with an alert and the true failure state recorded separately; the system never fabricates a new observation time and never fills a zero. This is the data-continuity guarantee of Section 9.1.

Two-level renormalization. The taxonomy acts at two levels. At the *factor* level, a unit with a missing factor is renormalized over its present factors within the block (a *Z*-state unit with no commerce coverage is measured on the other commerce factors). At the *block* level, a unit missing a whole block (e.g., no commerce data at all) is measured on the present blocks with the top-level weights renormalized. The two levels are applied in that order, and the result is a well-defined projection.

Proposition 6.4 (Two-level renormalization is a projection). *Let $w_1, \dots, w_K$ be positive block weights with $\sum w_k = 1$, and let a unit be present in a nonempty subset P of blocks. The two-level renormalized composite $\sum_{k \in P} \frac{w_k}{\sum_{j \in P} w_j} \hat{v}_k$ is the orthogonal projection of the full composite onto the subspace of present blocks, in the weighted ℓ_2 sense, and it lies in $[0, 1]$ whenever the present block values do.*

Proof. The weighted projection minimizes $\sum_k w_k (u_k - \hat{v}_k)^2$ over u supported on P ; the first-order conditions give $u_k = \hat{v}_k$ for $k \in P$ and the projection is exactly the re-normalized weighted mean. Convexity of the sum gives $[0, 1]$ containment. $\square$

The four-state taxonomy implements the design principle “a missing source never zeroes an Originator and never rewards absence.” The distinction between the four states is a judgment, and the judgment is registered (the classification of each factor for each unit is a versioned, auditable table, Section 10.2), so that a reader can in principle see *why* each missing value was handled the

way it was. This is the operational form of the missing-data discipline of Rubin [Rubin, 1976]: the mechanism of missingness is modeled explicitly rather than assumed away.

6.5 The FA composition

Blocks. Four blocks are computed per Originator:

$$\text{cap}_i = \sum_{k=1}^7 w_k \hat{x}_{i,k}, \quad \text{seven capability benchmarks, linear scale;} \quad (7)$$

$$\text{edge}_i = \frac{1}{4} \sum_{c=1}^4 s_{i,c}, \quad s_{i,c} = \frac{a_{i,c}^*}{\max_{j \in \mathcal{I}} a_{j,c}^*}, \quad \text{band-best capability per band;} \quad (8)$$

$$\text{eco}_i = \gamma_{\text{dev}} \text{dev}_i + \gamma_{\text{acad}} \text{acad}_i, \quad \text{developer stock and academic stock;} \quad (9)$$

$$\begin{aligned} \text{com}_i &= w_v \text{value}_i + w_p \text{price}_i \\ &\quad + w_f \text{free}_i + w_m \text{momentum}_i, \quad \text{commercial accessibility.} \end{aligned} \quad (10)$$

In (7), $\hat{x}_{i,k}$ are linear-normalized benchmark scores and w_k the source weights. In (8), $a_{i,c}^*$ is the capability of the unit’s best model in parameter band c (bands defined by total parameters; missing band contributes 0 and is always present in the mean — a true zero under Section 6.4, because a missing edge band is a fact about participation). In (9), the developer block is a heavy-tail mean over five ecosystem sources and the academic block is a mean of a heavy-tail citation ripple and a linear ledger of adjudicated artifacts: $\text{acad}_i = \delta_{\text{rip}} \text{ripple}_i + \delta_{\text{led}} \text{ledger}_i$. In (10), value is capability per unit blended price (with the same-model pairing rule — the capability score and the price must come from the same flagship model, so that “value” is not a category error mixing two different models), price is inverse price on a heavy-tail scale, free is boolean (true-zero discipline), and momentum is linearly normalized token adoption with a leaderboard-floor treatment for non-listed units.

Composite. The two vision groups (capability and ecosystem) are combined with group normalization, so that within-group block weights act within the group and top-level weights allocate between groups:

$$\text{cap}_i^* = \frac{\beta_{\text{cap}} \text{cap}_i + \beta_{\text{edg}} \text{edge}_i}{\alpha_{\text{cap}}}, \quad \alpha_{\text{cap}} = \beta_{\text{cap}} + \beta_{\text{edg}}, \quad (11)$$

$$\text{eco}_i^* = \frac{\beta_{\text{eco}} \text{eco}_i + \beta_{\text{com}} \text{com}_i}{\alpha_{\text{eco}}}, \quad \alpha_{\text{eco}} = \beta_{\text{eco}} + \beta_{\text{com}}, \quad (12)$$

$$\text{FA}_i = 100 (\alpha_{\text{cap}} \text{cap}_i^* + \alpha_{\text{eco}} \text{eco}_i^*). \quad (13)$$

The group-normalized form means the top-level weights allocate *between* the two vision groups while the block weights allocate *within* them. With the V1 values (Appendix B) the closed form is $\text{FA}_i = 100 (0.55 \text{cap}_i + 0.05 \text{edge}_i + 0.25 \text{eco}_i + 0.15 \text{com}_i)$, and the reading of FA is “percentage of the best-weighted combination in the included set.”

Proposition 6.5 (FA monotonicity in its blocks). *For fixed normalization extremes and fixed membership, FA_i is non-decreasing in each block $\text{cap}_i, \text{edge}_i, \text{eco}_i, \text{com}_i$ and in each within-block factor, and strictly increasing in any block whose weight is positive and whose value increases while all others are held fixed.*

Proof. Every block enters (13) through nonnegative weights, and each block is a nonnegative weighted mean of its factors, so the composite is a nonnegative linear combination of factor values. A nonnegative linear combination is non-decreasing in each argument, and strictly increasing when the coefficient is positive. $\square$

Monotonicity is the minimal sanity property a composite must have: it guarantees that no unit is *penalized* for improving on any measured dimension, so the composition is a proper score. It is worth stating because composite-indicator weighting schemes can violate it [Nardo et al., 2005] (e.g., through ratios in the denominator whose monotonicity is not checked); the production formula is verified to satisfy it by construction.

Adjustments. The settled anchor consumed by the index is

$$\text{FA}_i^{\text{adj}} = \text{FA}_i (1 + D_i) - R_i, \quad (14)$$

where $D_i \in [0, D_{\text{cap}}]$ is the sustained-leadership adjustment and $0 \leq R_i \leq R_{\text{cap}} \text{FA}_i$ the substantive-incident deduction (Appendix B). D is computed per board d with threshold set Y_d from the number of consecutive settled periods held within threshold y , $\text{streak}_{d,y}$:

$$D_{i,d} = \max_{y \in Y_d} w_y f(\text{streak}_{i,d,y}), \quad f(n) = \log_2(1 + n), \quad D_i = \sum_d \beta_d D_{i,d}, \quad (15)$$

with tier weights w_y and leaderboard weights β_d over the five ranked leaderboards (Table 19). Falling out of all thresholds on all boards sets D to zero immediately (no decay tail); re-entry accumulates from the current tier. The immediate reset is the double-counting discipline of Property 6.20: brand, mind-share, and lock-in inertia are already carried by the stock dimensions (capability and ecosystem slow variables), so a decay tail would count the same inertia twice. R applies only to officially confirmed substantive incidents, carries sources, and exposes a correction channel; because the incident may already have depressed the RI signal, the R settlement nets out the portion already counted through RI (one piece of evidence is never counted twice).

6.6 The RI renewal envelope

The live signal is an additive set of decaying pulses:

$$\text{RI}_i(t) = \sum_{e \in \mathcal{E}_i(t)} A_e 2^{-(t-t_e)/T_{1/2}(e)}, \quad (16)$$

where each event e has full amplitude A_e , start time t_e , and a half-life $T_{1/2}(e)$ from its content-type tier. Events are classified into four tiers (rank-driven, factual, opinion, emotional); tier half-lives are not disclosed.

Axiom 6.6 (State pricing). *The instantaneous value $\text{RI}_i(t)$ depends only on the current envelope state, not on the arrival path.*

The Axiom is enforced by the renewal rule: if a new event carries a full value $\geq$ the residual of the existing pulse for the same primary key (org, source, title), the old pulse is voided and the new full value starts a fresh pulse; otherwise the old pulse decays untouched. Deduplication by primary key prevents the same post from being counted twice across snapshots. Two consequences of state pricing are worth spelling out because they distinguish the design from a moving-average filter. *First*, the value of rank 1 is the same whether the unit climbed to it or jumped to it: the arrival path

is priced out of the signal by construction. *Second*, the renewal log (the list of voided pulses and their full values) is a deterministic replay ledger: given the log, any past instantaneous value can be recomputed exactly (Section 9.3).

Rank-driven amplitudes use an attention baseline: $A_e = \text{clip}(h_{\text{raw}}/h_{\text{ref}}, h_{\text{cap}}) w_{\text{src}} |p|$, with polarity $p \in [-1, 1]$; when no heat value is available the amplitude falls back to $\propto 1/\sqrt{\text{rank}}$. Factual posts use $A_e = B_{\text{base}} w_{\text{role}} w_{\text{kind}} p$, with an attention baseline (being discussed is itself weakly positive; only negatively factual signals are negative). During closed-market windows the signal only decays: no injection, no renewal, and settlement applies injected events at the next opening. The design deliberately does not use an exponentially weighted moving average; the renewal envelope is the system's answer to the cold-start and path-dependence defects of fixed-window exponential smoothing (Section 5.4).

Lemma 6.7 (Decay kernel positivity). *For $t \geq t_e$ and $T_{1/2}(e) > 0$, the kernel $2^{-(t-t_e)/T_{1/2}(e)}$ lies in $(0, 1]$ and is strictly decreasing in t .*

Proof. $t - t_e \geq 0$, so the exponent is ≤ 0 and the value lies in $(0, 1]$, with equality at $t = t_e$. Differentiating in t gives $-\ln 2 \cdot T_{1/2}(e)^{-1} \cdot 2^{-(t-t_e)/T_{1/2}(e)} < 0$. $\square$

Lemma 6.8 (Renewal jump magnitude). *Under the renewal rule, a single renewal jump of pulse e equals $A_e - r_e(t)$, where A_e is the new full value and $r_e(t)$ the residual of the voided pulse at the renewal time.*

Proof. Before the renewal the instantaneous contribution is the residual $r_e(t)$; after the renewal it is A_e (the new pulse starts at full value with $t_e = t$), and the renewal rule triggers exactly when $A_e \geq r_e(t)$, so the jump is nonnegative and equals $A_e - r_e(t)$. $\square$

6.7 The index family

The flagship composition (1) and the family surrounding it are:

$$\Theta_i(t) = \text{FA}_i^{\text{adj}}(t_0) + k_{\text{ri}}(\text{RI}_i(t) - \overline{\text{RI}}(t)) + \text{BR}_i(t), \quad \text{flagship actual}; \quad (17)$$

$$\overline{\text{RI}}(t) = \frac{1}{|\mathcal{I}(t)|} \sum_{j \in \mathcal{I}(t)} \text{RI}_j(t), \quad \text{included cross-sectional mean}; \quad (18)$$

$$\text{FA}_i^{\text{adj}}(t_0) \quad \text{settled anchor, frozen between settlements};$$

$$\text{RI}_i(t) \quad \text{live signal as a published series (code 8102)};$$

$$\text{H}_i(t) = \sum_{e \in \mathcal{E}_i(t)} |A_e| 2^{-(t-t_e)/T_{1/2}(e)}, \quad \text{absolute discussion heat}; \quad (19)$$

$$\Gamma_i = 1000 \frac{C_i}{C_{\text{frontier}}(t_{\text{base}})}, \quad \text{absolute capability composite}; \quad (20)$$

$$\Lambda_i(t) = C(t) \Theta_i(t), \quad \text{absolute composite, } C_0 = 1.0, \quad (21)$$

$$\mu_{\Gamma}(t) = \frac{1}{|\mathcal{I}(t)|} \sum_{i \in \mathcal{I}(t)} \Gamma_i, \quad \text{sector mean}; \quad (22)$$

$$M_{\Gamma}(t) = \max_{i \in \mathcal{I}(t)} \Gamma_i. \quad \text{sector maximum}; \quad (23)$$

In (20), C_i is the field-normalized capability level

$$C_i = \frac{\sum_k w_k \tilde{x}_{i,k} \mathbf{1}[i \in \mathcal{F}_k]}{\sum_k w_k \mathbf{1}[i \in \mathcal{F}_k]}, \quad \tilde{x}_{i,k} = \frac{x_{i,k}}{M_k}, \quad (24)$$

where $x_{i,k}$ is unit i 's raw capability datum on field k , M_k the field's normalization constant, $\mathcal{F}_k$ the set of units active in field k , and w_k the capability-source weights of (7); a unit is normalized over its active fields, so the definition is the two-level projection of Proposition 6.4 applied to the raw capability stock. Unlike the relative flagship, which uses included-set maxima at each settlement, the constants M_k are fixed across settlements; this is what lets Γ register simultaneous proportional progress by all units (the time-axis channel discussed below).

The family is organized by semantics, and the semantics are what the code-symbol binding in Appendix A (Table 11) records: Θ is the relative flagship (code 8100), FA^{adj} the transparent anchor (8101), RI the live signal (8102), H the absolute discussion heat (8111), Γ the absolute capability composite (8200), Λ the inflation-deflated absolute composite (8300), and μ_Γ, M_Γ the sector aggregates of Γ (9200). The regional variants (codes CN1111 , US1111 , CN3333 , US3333 , CN5555 , US5555) are country subsets: for each country, the Top-1 (1111), Top-3 (3333), and Top-5 (5555) composites over the country's Originators ordered by Θ . They exist so that a national audience can read its own landscape without losing the cross-sectional discipline of the full set.

The nominal line $\tilde{\Theta}_i(t)$ adds a zero-mean display-noise process to the actual line $\Theta_i(t)$; the process is deterministic pseudo-random (hash-seeded), has constant relative amplitude, and is forced to zero at the close of each window, so that the closing nominal equals the closing actual (Proposition 6.17). The absolute-heat line H is *pure actual*: it carries no display noise, because its volatility is itself the signal.

Two properties of the family deserve emphasis. *First*, the absolute family (Γ , μ_Γ , M_Γ) is what carries time-axis information that the relative flagship deliberately does not: because FA uses a cross-sectional maximum normalization, simultaneous proportional progress by all members leaves the relative index unchanged, and the absolute family is where such progress is visible (Proposition 6.3). *Second*, the deflated composite Λ multiplies the whole market by a common factor $C(t)$; it is the only family member that carries a slowly moving industry-level level, and because the factor is common, it preserves every cross-sectional ranking (Proposition 6.16). The two properties together implement the paper's central separation: relative measurement for fairness across units, absolute measurement for honesty about progress over time, never conflated.

6.8 The presentation layer as a controlled projection

Definition 6.2 (Presentation-layer mapping). A *presentation-layer mapping* is a function Π from the full indicator state $\xi(t)$ (Definition 6.1) to a public artifact $p \in \mathcal{P}$, where $\mathcal{P}$ is a small set of artifacts: a single number, a single line, and a consistent standard.

The presentation layer is *not* a simplification of the science; it is a well-defined, deterministic function of the state, and its fidelity boundary is the *communication anchor*: the artifact must preserve the ranking and the relative order of magnitude of the underlying state, must not reorder it, and must declare any aggregation it performs. Every simplification in the presentation layer must pass a “does not violate scientific fairness” check: it may omit detail, but it must not distort the ordering that the full state certifies. The honest claim for the reader of the artifact is *see the pattern, not master the technology*. This distinction is analyzed further in Section 7.4.

The presentation layer has a second, subtler job: it must present the same state honestly to two very different audiences at two different time scales. The *settlement audience* needs the exact, recomputable value (the actual line); the *intraday audience* needs a responsive, readable line without being misled into believing that micro-movements carry micro-signal. The dual-value process is the answer: the nominal line is the actual line plus a display-noise process that is (i) zero-mean, (ii) deterministic pseudo-random (hash-seeded, hence recomputable and auditable), (iii) bounded in

amplitude, and (iv) forced to zero at each close, so the settlement value is always the actual value. The display process is a presentation-layer concern and is parameterized in Appendix B (Table 22); it is never allowed to contaminate the actual line, and the actual line is the only line that feeds settlement and audit.

The presentation layer’s fidelity claim is therefore two-part and both parts are checkable: the mapping is a deterministic function of the state (a reader can in principle verify that the artifact is the declared function of the declared inputs), and the mapping preserves ordering and magnitude (the “does not violate scientific fairness” check of Section 8.5).

6.9 Invariants: propositions and proofs

The invariants below are the testable heart of the system. They are stated as propositions because each is a precise claim that can in principle be checked against public data, and they are proved because the system’s credibility rests on the reader being able to re-derive them rather than take them on faith.

Proposition 6.9 (Exact conservation of the adjusted pool). *For every time t ,*

$$\sum_{i \in \mathcal{I}(t)} \Theta_i(t) \equiv \sum_{i \in \mathcal{I}(t)} \text{FA}_i^{\text{adj}}(t_0). \quad (25)$$

Proof. Sum (17) over the included set. The centered term vanishes because $\sum_{i \in \mathcal{I}} (\text{RI}_i - \overline{\text{RI}}) = \sum_i \text{RI}_i - |\mathcal{I}| \cdot \frac{1}{|\mathcal{I}|} \sum_j \text{RI}_j = 0$, by the definition of the mean. The bridge slot BR is zero outside a member-set transition (Section 6.10), and when active is engineered to sum to zero over the included set at every time (Proposition 6.19). The result follows. $\square$

This is the mathematical mechanism behind “heat reorders positions but does not enlarge the total pool.” An arithmetic re-computation on a settlement tick confirmed the centered term vanishes to machine precision.

Corollary 6.10 (Zero-sum character of the live term). *The live term $k_{\text{ri}}(\text{RI}_i - \overline{\text{RI}})$ has zero sum over the included set at every time; it reallocates within the pool and never changes the pool total.*

Proof. Immediate from the proof of Proposition 6.9 and the definition of $\overline{\text{RI}}$. $\square$

Proposition 6.11 (Approximate conservation of the base pool). *Let D_{cap} and R_{cap} be the per-Originator caps on D and R , both relative to FA (Appendix B). Then*

$$\left| \frac{\sum_{i \in \mathcal{I}} \Theta_i - \sum_{i \in \mathcal{I}} \text{FA}_i}{\sum_{i \in \mathcal{I}} \text{FA}_i} \right| \leq D_{\text{cap}} + R_{\text{cap}}. \quad (26)$$

In the special case where no Originator holds a leadership streak ($D \equiv 0$) and no substantive incident is recorded ($R \equiv 0$), the sums are equal.

Proof. From (14), $\text{FA}_i^{\text{adj}} = \text{FA}_i(1 + D_i) - R_i$. Summing over the included set and using Proposition 6.9, $\sum \Theta = \sum \text{FA} + \sum \text{FAD} - \sum R$. Since $0 \leq D_i \leq D_{\text{cap}}$ and $0 \leq R_i \leq R_{\text{cap}} \text{FA}_i$, the per-Originator deviation from base FA is bounded by $D_{\text{cap}} \text{FA}_i + R_{\text{cap}} \text{FA}_i$; dividing by $\sum \text{FA}$ gives (26). When $D \equiv 0$ and $R \equiv 0$, the deviation vanishes. $\square$

The “approximately” in the paper’s conservation claim is this proposition, and it is stated this precisely because the published index carries the adjustments. A reader who checks the conservation against the documented FA methodology and the published snapshot (Appendix G) is checking *exact* conservation against FA^{adj} and *approximate* conservation against FA, with the bound given here.

Proposition 6.12 (Well-posedness and boundedness). *Under the unified scale contract (3)–(6) and the missing-value taxonomy (Section 6.4), every factor value and every composite value lies in $[0, 1]$; consequently $\text{FA}_i \in [0, 100]$ for all i .*

Proof. By induction over the composition tree. Linear and heavy-tail normalizations map to $[0, 1]$ (the maximum is attained in $\mathcal{I}$; $\log_{10}$ is monotone with $\log 1 = 0$); the inverse rule preserves non-negativity before normalizing; boolean factors are in $\{0, 1\}$; the missing-value protocol assigns either 0 (true zero, weight-preserved), a re-normalized presence, or a leaderboard floor in $[0, 1]$. Weighted arithmetic means of values in $[0, 1]$ stay in $[0, 1]$. $\square$

Boundedness is not a technicality; it is what makes the scale *interpretable* (as “percentage of the best-weighted combination”) and what makes the audit inequalities of the live signal meaningful. A composite that could take arbitrary values would carry no such semantics.

Remark 6.2 (Pool boundedness and the balance reading). *The boundedness of Proposition 6.12 is per-Originator; it extends at once to the pool. Write $n = |\mathcal{I}|$ for the membership size, and $S_b = \sum_{i \in \mathcal{I}} b_i$ for each block $b \in \{\text{cap}, \text{edge}, \text{eco}, \text{com}\}$. Under the scale contract (linear (3), heavy-tail (4), and boolean (6) forms, with the inverse and floor conventions used in the commerce block), every normalized factor value lies in $[0, 1]$ and the value 1.0 is attained by at least one included unit, so each factor’s column sum lies in $[1, n]$. Each block is a weighted mean of its factors, so $S_b \in [1, n]$. Writing the closed form (13) with block weights $\beta_{\text{cap}}, \beta_{\text{edge}}, \beta_{\text{eco}}, \beta_{\text{com}}$ — summing to one and fixed in Appendix B — as $\text{FA}_i = 100 \sum_b \beta_b b_i$ and summing over $\mathcal{I}$ gives*

$$\sum_{i \in \mathcal{I}} \text{FA}_i = 100 \sum_b \beta_b S_b \in [100, 100n]. \quad (27)$$

The lower bound is attained when a single unit leads every factor; the upper bound when every included unit attains the leader’s value on every factor.

An equivalent form exposes the frontier. Let $\hat{x}_{i,f} \in [0, 1]$ be the normalized value of factor f for Originator i , and α_f its composite weight (block weight times within-block weight, so $\sum_f \alpha_f = 1$). Then

$$\sum_{i \in \mathcal{I}} \text{FA}_i = 100 (n - G), \quad G = \sum_f \alpha_f \sum_{i \in \mathcal{I}} (1 - \hat{x}_{i,f}). \quad (28)$$

Since for every factor at least one included unit attains 1.0, $G \in [0, n-1]$: the pool is a pooled distance from the composite frontier. For the current membership ($n = 10$) the bounds read $[100, 1000]$, i.e. $\sum_{i \in \mathcal{I}} \text{FA}_i = 1000 - 100 G$ with $G \in [0, 9]$.

Read as a share, the pool over $100n$ is the fraction of the composite frontier jointly held. A single-actor monopoly driving every factor to its own maximum pushes the pool toward 100; a perfectly competitive field bunched at the frontier pushes it toward $100n$. The pool is thus an industry-balance indicator; the current instance is reported with the snapshot (Appendix G).

By Proposition 6.9, the Eins Index pool equals the adjusted-anchor pool; when the D adjustment vanishes ($D \equiv 0$, $R \equiv 0$) this equals the FA pool exactly, and Proposition 6.11 bounds the relative gap otherwise. The balance reading therefore transfers to the Index.

The ceiling scales as $100|\mathcal{I}|$, so the membership-invariant reading is the share of the maximal pool, not the absolute ceiling.

Proposition 6.13 (State-pricing consequences). *Under Axiom 6.6 and the renewal rule, (a) a single-step renewal jump equals full value minus residual; (b) a fresh event enters at full value with zero response lag; (c) at any instant at most one active pulse exists per primary key (org, source, title), and the renewal log is a deterministic replay ledger.*

Proof. (a) The instantaneous value before the renewal is the residual r ; after the renewal it is the full value A , by the rule that $A \geq r$ triggers replacement; the jump is $A - r$ (Lemma 6.8). (b) A fresh pulse has $t_e = t$ so the exponential factor is 1 (Lemma 6.7). (c) The renewal rule modifies in place and never appends a second pulse for the same key; the log therefore replays deterministically. $\square$

Proposition 6.14 (Audit inequality). *For every Originator i and time t , over the same envelope set,*

$$H_i(t) \geq |RI_i(t)|. \quad (29)$$

Proof. Both sums run over the same active set with the same kernels. By the triangle inequality, $|\sum_e A_e w_e| \leq \sum_e |A_e| w_e$ with $w_e = 2^{-(t-t_e)/T_{1/2}(e)} > 0$, which is exactly $|RI| \leq H$. $\square$

This inequality is the testable invariant behind the dual-track audit: the absolute-heat line is always at least as large in magnitude as the signed live line, and any violation is a data-integrity alarm. Because H and RI are computed from the same envelope set by formulas that differ only in the absolute value, the audit is a strong check — it is not a bound that could be satisfied vacuously; it holds tick by tick and its failure is diagnostic (Section 9.3).

Proposition 6.15 (Monotonicity of sector aggregates, conditional). *Holding membership, scale, and method fixed, if every member's Γ is non-decreasing over a window, then the sector mean μ_Γ and the sector maximum M_Γ ((23)) are non-decreasing over that window.*

Proof. The mean is a positive linear combination of the members' Γ ; the maximum is an order-preserving map. Both are non-decreasing functions of non-decreasing arguments. The premise fails whenever membership changes, which is a real constituency change rather than a method artifact; the sector aggregates do not claim unconditional monotonicity (Section 6.10). $\square$

Proposition 6.16 (Rank invariance under the deflator). *Multiplication by a common positive deflator $C(t)$ as in (21) preserves the cross-sectional ranking and all relative gaps of Θ at every time.*

Proof. $C(t)$ factors out of every ratio and every comparison: for i, j , $\frac{\Lambda_i}{\Lambda_j} = \frac{\Theta_i}{\Theta_j}$ and $\Lambda_i > \Lambda_j \iff \Theta_i > \Theta_j$. $\square$

Proposition 6.17 (Nominal display is drift-free and converges). *The display-noise process $x(t)$ has zero mean and is forced to zero at the close of each window; the nominal line therefore does not drift systematically from the actual line, and at the close nominal $\equiv$ actual.*

Proof. $x(t)$ is a deterministic pseudo-random, hash-seeded discrete Ornstein–Uhlenbeck process [Uhlenbeck and Ornstein, 1930] with zero-mean increments, so $\mathbb{E}[x(t)] = 0$ by construction. The taper window multiplies both new innovations and existing stock by $(t_{\text{close}} - t)/N_{\text{taper}}$, driving $x \rightarrow 0$ as $t \rightarrow t_{\text{close}}$; the deterministic seeding makes the entire nominal series recomputable. $\square$

The OU process is used in its classical role as the standard model of a mean-reverting, bounded random walk; the innovation here is not the process but its discipline: deterministic seeding (recomputable), zero mean (drift-free), and forced close convergence (settlement integrity).

Proposition 6.18 (Centering preserves cross-sectional ordering of the anchor). *For two Originators i, j with equal live signals $RI_i(t) = RI_j(t)$, the flagship ordering $\Theta_i(t) \geq \Theta_j(t)$ is determined entirely by the anchor ordering $FA_i^{\text{adj}}(t_0) \geq FA_j^{\text{adj}}(t_0)$; the live term cannot flip it.*

Proof. If $RI_i = RI_j$ then the centered terms are equal (both equal $k_{ri}(RI_i - \overline{RI})$), and with BR zero the difference $\Theta_i - \Theta_j = FA_i^{\text{adj}} - FA_j^{\text{adj}}$. $\square$

The proposition states the boundary of the live signal’s authority: discussion can reorder units with *different* live signals, but among units with the same live signal the anchor’s ordering is preserved exactly. The live signal is a deviation mechanism, not a certification mechanism — the same boundary claim as Section 5.2, now at the level of a theorem.

6.10 The BR transition

The bridge slot $BR_i(t)$ in (1) exists for exactly one purpose: to carry an Originator across a *compositional transition* of the reference set without letting the transition’s mechanical arithmetic masquerade as a change in the measured quantity. The first and currently only confirmed type is the member-set transition BR-01.

Let I^- be the unit’s formally published value before a member-set change, I^+ its fair value under the new membership with the new FA, and I^{mech} the counterfactual value that would have been published if the new membership were in force while the old FA were frozen. Define the mechanical and fundamental changes as

$$G_{\text{mech}} = I^{\text{mech}} - I^-, \quad \Delta_{\text{fund}} = I^+ - I^{\text{mech}}. \quad (30)$$

The bridge is initialized to cancel exactly the mechanical part and reveal exactly the fundamental part:

$$BR(0) = -G_{\text{mech}}, \quad I^{\text{pub}}(0) = I^+ + BR(0) = I^- + \Delta_{\text{fund}}. \quad (31)$$

Thereafter the bridge decays along a deterministic path to zero:

$$BR(t) = -G_{\text{mech}} \varphi(t), \quad \varphi(0) = 1, \quad \varphi(H) = 0, \quad (32)$$

so the published path starts at $I^- + \Delta_{\text{fund}}$ and converges to the new fair value I^+ .

Proposition 6.19 (BR transition convergence). *Under (32) with φ continuous and $\varphi(H) = 0$, (a) the bridge is zero at time H and thereafter; (b) the published path converges to the new fair value I^+ ; (c) the bridge never deposits into FA and never enters the RI feed; and (d) if $G_{\text{mech}} = 0$ (no mechanical change), no bridge is generated.*

Proof. (a) By the terminal condition $\varphi(H) = 0$. (b) Since $BR(t) \rightarrow 0$, $I^{\text{pub}}(t) \rightarrow I^+$; the initial value is $I^- + \Delta_{\text{fund}}$ which need not equal I^+ , so the path is finite-duration interpolation, and the terminal value is exact. (c) The bridge is a display-path device; the FA anchor is defined on the new membership independently, and the RI feed only accepts event pulses, which the bridge never creates. (d) $BR(0) = -G_{\text{mech}} = 0$, and the zero bridge stays zero. $\square$

The engineering meaning of the proposition is the separation it enforces: a purely mechanical revaluation caused by the roster changing (new member redefines a normalization extreme, shifting everyone’s level) is *temporarily* bridged so that the published path does not jump for an arithmetic reason, while the genuine fundamental change (Δ_{fund}) is revealed immediately and the published path terminates at the new fair value. The bridge is bounded, terminates, and never contaminates the anchor or the live feed. When the counterfactual I^{mech} cannot be reliably reconstructed, no un-audited bridge is generated (Section 10.5); the sector aggregates μ_Γ, M_Γ deliberately do not use BR-01, because a membership change alters the statistical population they describe and that change is real (Proposition 6.15).

Property 6.20 (Zero-reset of sustained leadership). *Dropping out of all leadership thresholds on all boards sets $D = 0$ immediately, with no decay tail; re-entry accumulates from the current tier. This prevents double-counting: brand, mind-share, and lock-in inertia are already carried by the stock dimensions (capability and ecosystem slow variables), so a decay tail would count the same inertia twice.*

6.11 Discussion of design choices

The invariants above are deliberate design choices; we state which and why.

Why a centered composition? The conservation law (Propositions 6.9–6.11) is what turns a set of relative scores into a *scale* with a stable center: attention can reorder positions but cannot, in aggregate, manufacture total influence. This is the property that the paper identifies as the measurement-scale contribution of the system. The cost is that total-pool information is not carried by the flagship line; it is carried instead by the absolute family, which is the right place for it (Section 6.7).

Why two temporally separated components? FA is a stock with no time decay; RI is a flow with mandatory decay. The separation is what keeps the anchor scrutable (slow, source-rich, public) while the live signal stays responsive (but manipulable, hence opaque). The two components answer two different questions — “what is the cumulative footprint?” and “what is happening right now?” — and forcing them into one homogeneous score would destroy both answers.

Why the missing-value taxonomy is four-state, not binary? A binary “present/absent” collapses three genuinely different situations — the unit participates with zero activity (true zero), the source does not cover the unit (numerical absence), and the unit fell below the printed cutoff (leaderboard floor) — plus the infrastructural case (technical unavailability). Treating them identically would either reward absence (erasing true zeros) or punish it (zeroing a covered-but-unlisted unit); the four-state taxonomy is the minimal classification that avoids both errors, and its cost — judgment calls — is managed by registering every classification (Section 6.4).

Tension with Goodhart. Any index can be gamed (Sections 1.2 and 7.5). The design does not pretend to eliminate gaming; it confines it. The conservation law bounds the aggregate damage of gaming the live signal; source diversity and replaceability raise the cost of gaming the anchor; the volatility budget bounds the damage of a single manipulated burst. These are engineering responses to a formal threat model, analyzed in Section 7.5.

Part V

Trust and Verification

7 Trustworthiness: Why the Index Can Be Scrutinized

7.1 Scientific

The four elements of a measurement scale (Section 5.5) are met: a common scale (the unified contract (3)–(6) over a shared reference set); a stable center (the conservation law); domain-universal operations (every unit passes through the same composition tree); and comparability across time (settlement windows and a frozen anchor). The claims that can be checked against public data are stated as propositions (Section 6.9) and are partially reproducible: the conservation and boundedness invariants can in principle be recomputed from the documented FA methodology, the reference parameters, and the published snapshot (Appendix G). The system is *designed to remain scrutable even where it is not end-to-end reproducible*: the algebraic structure of the withheld components is public, and only their numerical calibration is reserved.

The honest boundaries of the scientific claim are: the board is small (twelve Originators in the first assessment); confidence intervals from overlapping capability sources overlap substantially for mid-board units, and ties are reported transparently rather than broken arbitrarily; and the conservation claim on the *base* anchor is approximate (Proposition 6.11). None of these boundaries

is hidden; Section 8.9 states them in full.

The scientific claim also has a negative form that is as important as the positive one: the system is designed so that its *invariants* are falsifiable. If an independent recomputation of the conservation finds it violated, or of the audit inequality finds it broken, the system has failed a test it itself declared. This is the difference between a measurement system and a dashboard: a dashboard can be wrong in ways that are invisible by construction, while a measurement system with stated invariants can be wrong in ways that are *publicly detectable*. That is the sense in which “scrutable” is a load-bearing property.

7.2 Fair

Fairness is designed as an institutional property, not a numerical one. Three mechanisms carry it.

Asymmetric disclosure. The system opens what is hard to manipulate and closes what is easy to manipulate. FA inputs are slow stocks from third-party sources; its methodology and reference parameters are documented here. RI is a fast, short-horizon sentiment signal; its per-source weights, half-life tiers, baselines, and implementation are retained. The institutional reason is manipulation resistance: a rating or index that publishes exact short-horizon parameters invites arbitrage against the measure, a well-known failure mode in evaluation practice [Manheim and Garrabrant, 2018]. This is the same posture as index and rating providers whose methods are public while exact parameters are reserved.

Don’t measure yourself with your own ruler. The FA backbone is third-party open data; self-collected public data is a supplement and is reconcilable against the third-party sources. Sources are replaceable: each source has an independent circuit breaker, multiple sources are weighted for the same quantity, and a source-swap migration procedure exists (Appendix D). The system is therefore structurally *not in competition with its sources* — it does not produce, own, or sell the underlying data (Section 13) — and this is the institutional guarantee that it has no incentive to distort a source’s signal. The argument is structural, not hortatory: the decoupling of the method layer from the data layer (Section 13.1) means the system’s own incentives point toward source fidelity, not source manipulation.

Bounded negative correction. The real-harm correction R applies only to officially confirmed substantive incidents (data breaches, critical security vulnerabilities, regulatory penalties), with sources attached and a correction channel open; negative *attention* never enters FA through any other path. The system declares itself an *independent third-party evaluation, not an endorsement*, and does not use the logos of the measured organizations, in respect of corporate-reputation rights. The boundedness of R (a small fraction of FA, Appendix B) is a fairness design choice: the anchor is a measure of cumulative influence, and a single incident — however serious — is not allowed to erase a cumulative record. The RI negative tracks and the risk family carry the incident’s current-period impact; the anchor carries a bounded, verified correction.

7.3 Real-time

“Real-time” is an engineering caliber, and we separate the three layers in which it is realized: *collection* (sources polled at L1 cadence, on the order of 1–5 minutes), *pricing* (the composition engine at L2, bucketed), and *display* (the flagship line published at minute level, with second-level display capability). We distinguish the weak claim (*updates quickly*) from the strong claim we make (*a market-semantics quote machine*): a real-time sequence, periodic review-and-release of the slow anchor, and a suspend–resume release mechanism when measurement integrity is at risk. During closed-market windows the anchor is frozen and the live signal only decays; reopenings are announced

and settled in batch. This is the research wording of what a market practitioner would call “quote / close / halt / resume”.

The real-time claim is deliberately scoped. It is a claim about the *pipeline* (collection at L1 cadence, pricing at L2, display at minute level), not a claim that the underlying construct moves at that speed. The anchor is a slow variable and moves at settlement; the live signal moves at the pipeline cadence; the display shows both. Confusing “the display updates every minute” with “influence changes every minute” would be a category error, and the dual-value process (Section 6.8) is designed to make the distinction legible.

7.4 Accessible: a property of the index layer

The paper’s premise about accessibility is narrow and precise: accessibility is a property of the *index output*, not of the research system, and not of this document. The presentation-layer mapping (Definition 6.2) is the formal object that makes “accessible” checkable: it is a deterministic projection of the full indicator state onto a single number, a single line, and a consistent standard, with a declared fidelity boundary (the communication anchor of Section 6.8). The mapping exists so that a layperson can read the pattern without mastering the technology — *see the pattern, not master the technology*. Every simplification in the output is subject to the same “does not violate scientific fairness” check as the presentation layer itself. The V1 snapshot, an “understand at a glance” demonstration (Appendix G), exercises the output layer only.

7.5 Reflexivity and manipulation resistance

We treat manipulation as a formal threat model rather than a public-relations concern. An attacker who controls a manipulable input can push the index toward a target value; the design’s response is layered. (i) The conservation law (Propositions 6.9–6.11) bounds the aggregate effect: a burst that inflates one Originator’s live signal must, by the centering, deflate the others, and the total pool is anchored by FA. (ii) The volatility budget bounds the single-burst magnitude; a per-Originator movement bound raises an audit alert without truncating the value. (iii) The anchor is hard to game because it is a weighted mean of independent third-party stocks whose replaceability is institutional (Section 7.2). (iv) The asymmetry of disclosure denies an attacker the exact short-horizon parameters that would make the live signal predictable. This is the engineering reading of Goodhart-style failure [Manheim and Garrabrant, 2018, Goodhart, 1984]: the measure is designed so that its target is costly to hit.

Reflexivity deserves a separate paragraph because it is the subtle case. A public influence index does not merely measure; it changes the measured, because the measured actors can observe it and adjust. The reflexive loop “discussion of the index feeds the index” is real — self-fulfilling dynamics of public ratings are documented in experimental cultural markets [Salganik and Watts, 2008] — and the system’s response is deliberately conservative: the reflexive amplifier (a mechanism that would feed index-related attention back into the index) is disabled in V1, and index-related discussion enters the live feed only through the ordinary channels that any entity-word discussion uses — with the same source verification, the same polarity model, and the same budget. Whether this is sufficient is an empirical question, and the reflexivity of the system is itself listed as a study object (Section 12.3). The design’s position is that reflexivity cannot be eliminated, only made legible and bounded.

8 Verification: Segment-by-Segment Acceptance and Evidence

8.1 Acceptance framework

We verify the three scientific properties (*scientific*, *fair*, *real-time*) against acceptance criteria stated *before* the evidence. The accessibility property is verified against the fidelity-boundary criteria of the presentation layer (Section 6.8). The construct-validity property is verified against the acceptance criteria of Section 8.6. Evidence is labeled by strength: where the window is short, evidence is *illustrative*, not statistical.

8.2 Scientific acceptance

Criterion: the invariants in Section 6.9 hold on the published data, and the scale behaves as a slow-anchor / live-signal split.

Evidence. The conservation identity was confirmed on a settlement tick (Proposition 6.9); the audit inequality (Proposition 6.14) holds per-tick over the observation window. The separation of time scales is visible in the semantics of the two components: FA is a monotone stock, RI a decaying flow; over an observation window as short as the current one the anchor is expected to move little while the live signal moves much more. A quantitative confirmation of this time-scale split requires a longer window and is listed as future work (Section 8.9). The FA snapshot that supports this discussion is given in Appendix G with its collection date; it is a snapshot, not a time series, by design (the live signal is not reproduced here).

Counterfactual that would fail the test. A violation of the conservation identity on a settlement tick, or a violation of the audit inequality on any tick, would fail the criterion. Neither occurred over the window. Because the window is short, the time-scale-separation evidence is illustrative, and the claims are invariant checks and design properties, not hypothesis tests [Amrhein et al., 2019].

8.3 Fairness acceptance

Criterion: reconciliation is possible, source coverage is disclosed, corrections are logged, and the measured inputs are third-party-led.

Evidence. The dual-track audit (the inequality of Proposition 6.14) was used as an automatic reconciliation check between the absolute-heat and signed lines across the window. Source coverage and the missing-value protocol are documented (Appendix B); a correction log records adjustments to the ledger. The organization and tool attributions in the first scope (Section 3.6, Appendix C) were checked item-by-item against primary sources at writing time to the extent web verification was available. **Counterfactual that would fail the test.** A missing-value handling that could not be reconciled against its registered classification, or a ledger correction with no log entry, would fail the criterion. The audit trail is designed so that the failure would be detectable in principle.

8.4 Real-time acceptance

Criterion: the three-layer latency calibers (collection / pricing / display) are measured, and the response to a known event is demonstrable.

Evidence. The pipeline is run at the calibers of Section 7.3: collection on the order of 1–5 minutes, pricing at L2 bucketing, display at minute level. Because the window is short and the live signal is not reproduced here, the event-response demonstration is *illustrative* rather than a statistical test: the settlement-tick behavior (frozen anchor, batch injection, centering) and the

reopen semantics were observed to match the design. A full latency and event-response study is future work.

8.5 Accessibility acceptance (index layer)

Criterion: the presentation-layer artifacts (a number, a line, a standard) are produced by a well-defined mapping that preserves ranking and relative magnitude, and pass the “does not violate scientific fairness” check.

Evidence. The mapping is the projection of Section 6.8; by construction it preserves the ranking (it is order-preserving on the flagship line) and declares its aggregations. The V1 snapshot in Appendix G serves as the “understand at a glance” demonstration of the output layer only. This is evidence about the presentation layer’s fidelity, not scientific evidence about the world.

8.6 Construct validity acceptance

The construct-validity claims of Section 4 need their own acceptance framework, because they are claims about a theoretical construct rather than about the system’s internal consistency. Table 5 states, for each validity claim, the acceptance criterion, the evidence status, and the failure condition.

Table 5: Construct-validity acceptance criteria. “Evidence status” uses **now** (evidence present at writing), **planned** (design defined, data not yet collected), and **running** (continuous monitoring).

Claim	Acceptance criterion	Status	Failure condition
Content validity	every block/track maps to a definition channel; no channel empty	now (Prop. 4.1)	a block with no channel, or a channel with no block
Convergent, channel	independent evaluation suites agree with in-block ordering	planned	disagreement beyond stated tolerance across methods
Convergent, external	channel scores correlate with non-input adoption measures	planned	near-zero or sign-reversed correlation
Discriminant, channels	capability vs. ecosystem rankings not near-perfectly collinear	now (§11.2)	collinearity consistent with a single latent factor
Discriminant, time	FA stable within window; RI moves	planned	comparable movement on both scales
Known-groups	top-tier Originators are documented-ubiquitous	now (weak)	a no-evidence unit ranked above a documented-ubiquitous one
Event-consequence	release/incident moves predicted components	planned	systematic sign reversal on events

The framework is the paper’s answer to “how will anyone know the index measures influence?” — not by the system’s assertion, but by a running program of tests whose failure conditions are stated in advance. The distinction between **now**, **planned**, and **running** is the honesty of the framework: at writing, the system has internal-consistency evidence and preliminary discriminant evidence, and is missing the external convergent evidence that only time and independent data can supply.

8.7 Quantitative FA sensitivity design

Composite indicators are only as trustworthy as their sensitivity analysis; the methodological literature is unambiguous that an indicator whose ranking flips under small weight perturbations is not a measurement, it is a choice [Saisana et al., 2005, Nardo et al., 2005]. The system therefore adopts a quantitative sensitivity design *before* the V1 release is treated as authoritative. The design follows the uncertainty and sensitivity framework of Saisana et al. [Saisana et al., 2005]: perturb inputs and weights over stated ranges, and report how much the rankings and the invariants move.

Table 6: Quantitative FA sensitivity design. Each factor is a free parameter in the planned experiment; “perturbation” states the design range.

Factor	Baseline (V1)	Perturbation	Reported output
Top-level weights	0.60/0.40	± 0.10 (composition)	rank flips
Within-group weights	0.55/0.05; 0.25/0.15	± 0.05	rank flips
Capability source weights	7 sources	$\pm 30\%$ each	rank flips
Missing-value classification	registered table	re-classify boundary cases	rank flips
Normalization rule	linear / heavy-tail	swap per factor class	rank flips
Source coverage	current roster coverage	drop one source at a time	rank flips; conservation
D , R caps	Appendix B	0 to cap	conservation bound

The two reportables are (i) the *rank-flip matrix* (how many pairwise rankings reverse under each perturbation, summarized by a flip count and the largest single-factor flip) and (ii) the *invariant stability* (conservation and audit inequalities under perturbation, which must hold by construction and are re-checked numerically). The design’s acceptance criterion is stated in advance: a factor whose removal or re-weighting flips more than a stated fraction of the cross-sectional rankings is flagged as a *dominant factor*, and the committee decides whether the factor’s dominance is a fact about the field or a defect of the composition (Section 10). The sensitivity analysis also covers the Leamer-style *extreme-bounds* question [Leamer, 1983]: report the range of each unit’s rank as weights range over a plausible box, so that a reader sees not one ranking but the *robustness region* of the ranking.

8.8 Failure modes and threat matrix

Manipulation resistance (Section 7.5) and data integrity (Section 9.1) are formal enough to deserve a consolidated threat matrix. Table 7 lists the failure modes the design defends against, the mechanism of the failure, the defense, and the residual risk that remains after the defense. The residual risks are the honest remainder; they are the input to the governance and monitoring agenda (Section 10).

Twelve rows is not a complete threat catalog; it is the catalog the design is *engineered against*, and it is published so that the residual risks are visible to the same scrutiny as the defenses. The governance agenda of Section 10 is organized around the residual column: every residual risk has an owner (an audit check, a monitoring loop, or a governance review).

8.9 Limitations

Because FA is a slow-moving anchor by design (Section 5.3), the first point-in-time snapshot is a meaningful reading on its own; claims about how the anchor moves between successive settlements await a longer window and are deferred. The board is small (twelve Originators in the first assessment), so relative measures are coarser than they will be after expansion. Overlapping capability sources

Table 7: Failure-mode and threat matrix. “Defense” is the design mechanism; “residual” is what remains even with the defense in place.

Failure mode	Mechanism	Defense	Residual risk
Benchmark contamination	leaked items inflate capability	multi-source weighted capability; contamination-aware sources	slow contamination across all sources
Assessor bias	LLM-as-judge self-preference	human adjudication of polarity; value exposure	residual annotator variance
Rank gaming	fake popularity moves the live signal	centering; budget; verified sources; opaque params	coordinated cross-source bursts
Source capture	one provider controls a factor	circuit breakers; redundancy; replaceability	slow drift in the source itself
Attribution error	artifacts mapped to wrong Originator	registered mapping; unmapped bucket	legacy misattribution in history
Missing-data gaming	absence converted to advantage	four-state taxonomy; registered classifications	boundary-class misjudgment
Total-pool inflation	uncentered live term enlarges pool	centering (Prop. 6.9)	residual approximation from D, R
Reflexive amplification	index discussion feeds the index	amplifier off in V1; ordinary-channel treatment	undetected indirect reflexivity
Display misreading	micro-movement read as micro-signal	dual-value process; close convergence	residual misreading of noise
Suspend evasion	halt used to hide bad news	halt limited to integrity events; disclosed	misjudged halt boundaries
Weight capture	committee captured by measured actors	published rules; version vectors; independent sources	slow institutional capture
Method myopia	wrong construct operationalized	construct-validity program	unfalsified wrong construct

create correlated mid-board noise; ties are reported transparently. The unified calibration experiment — a batch re-estimation of the V1 reference parameters against held-out evaluation — is future work (Section 12.2); the released values are V1 production values, not a placeholder awaiting calibration. RI is strongly correlated with events and times, and its numerical realization cannot be reproduced by an external party by design; this is a boundary of the measurement (its live component is not end-to-end verifiable) rather than a hidden defect, and it is the price of manipulation resistance (Section 7.5). Two further boundaries are stated for completeness. First, the external convergent evidence for construct validity is planned, not yet collected; the index is a measurement system whose validity case is in progress (Section 8.6). Second, the composition’s numerical weights are V1 values whose robustness region is not yet measured; the sensitivity design of Section 8.7 is the plan, and the weights are production values, not a calibrated optimum.

Part VI

Operations and Evidence

9 Data Dictionary, Pipeline, and Audit

9.1 Data dictionary and quality states

Measurement systems live or die by their data contract. Table 8 is the system’s data dictionary at the level of source families and quality states; the full registry is versioned in the engineering system. Two design properties of the dictionary deserve emphasis before the table. *First*, every field carries its observation time, its quality state, and its source and method version — a value without its provenance is not a measurement, it is a claim. *Second*, every field has an explicit *non-observation semantics*: a missing or failed field is recorded as a failure state, never as a silently fabricated zero, and the most recent valid value continues with an alert (Section 6.4).

Table 8: Data dictionary: source families, cadence classes, and the quality states that gate each observation. “Cadence class”: fast (minutes), daily, event.

Source family	Measured content	Cadence class	Quality states
Capability bench- marks (7 sources)	model capability scores, crowd ratings	daily/slower	FRESH / LAGGING / STALE / EXPIRED
Open-source ecosys- tem	repositories, downloads, likes, models	fast/daily	FRESH / DEGRADED / QUARANTINED
Academic ledger	curated artifacts, cita- tion ripple	event	VERIFIED / PENDING / RE- JECTED
Commercial routing	API prices, official free API access, token adop- tion	daily	FRESH / PARTIAL / MISS- ING
Public-platform sig- nals	ranking positions, heat, authoritative posts	fast	OBSERVED / MISS- ING / DEGRADED / NO_OBSERVATION
Adjudicated events	incident records, release events	event	CONFIRMED / PENDING / UNRESOLVED

The quality-state vocabulary is deliberately small and shared across families, because the downstream logic (circuit breakers, missing-value classification, audit) is state-driven: each state has a defined effect on how the observation enters (or does not enter) the computation. The mapping from raw state to effect is registered and versioned, so that “why did this value not update?” has a mechanical answer (Section 9.3).

9.2 Collection–pricing–display pipeline

The pipeline that turns external reality into a published line has three stages, matching the three layers of Section 6.1 and the three latency calibers of Section 7.3:

Collection (L1). Fast sources are polled on the order of 1–5 minutes; slow and event sources on daily and event cadences. Raw observations are written to an append-only store with their observation times and source versions. The collection stage never interprets; it records.

Pricing (L2). The derived-indicator registry computes normalized factors, blocks, the settled anchor, and the RI envelope tracks. This stage is deterministic: the same L1 observations and versions produce the same L2 output. It is the stage where the missing-value taxonomy and circuit breakers act.

Display (L3). The index family is computed and projected through the presentation-layer mapping (Section 6.8). This is the stage where the actual and nominal lines diverge and reconverge at each close.

Two pipeline disciplines deserve emphasis. *Idempotency*: every stage is designed to be recomputable from its inputs and versions, so the pipeline can in principle be replayed from any point (Section 9.3). *No retroactive editing*: the display history is written once; corrections propagate as new versions, never as edits to old ones (the no-backcasting discipline of Section 3.5, applied at the data level). A reader of the published series can therefore in principle tell *what was believed when*, which is the property designed to make the series auditable rather than merely available.

9.3 Audit and replay protocols

The system’s audit protocol is built from the invariants of Section 6.9 plus the replay property of the layers. It runs continuously and reports four checks:

1. **Dual-track audit.** The audit inequality $H_i(t) \geq |RI_i(t)|$ (Proposition 6.14) is checked per tick; a violation is a data-integrity alarm. Because both lines derive from the same envelope set by formulas differing only in the absolute value, the check is tight: it detects source desynchronization, misattribution, and envelope corruption.
2. **Conservation audit.** The flagship conservation (Proposition 6.9) is rechecked at every settlement; the bound approximation (Proposition 6.11) is rechecked whenever D or R is nonzero.
3. **Replay audit.** The RI renewal log is a deterministic ledger (Lemma 6.8, Proposition 6.13): any past instantaneous value is recomputed exactly from the log; the audit is designed so that the published live line can be checked against the log’s arithmetic image.
4. **Reconciliation audit.** The self-collected observations are reconciled against third-party sources where both exist; a persistent unreconciled gap is flagged and investigated (Section 9.4).

The replay protocol is the most distinctive element: it means the live signal is *auditable without being reproducible*. An external party cannot reproduce the RI series from scratch (the parameters are reserved), but an auditor granted the renewal log can in principle verify that the published series matches what the log implies. Auditability and non-reproducibility are not in tension; they are separated by the log.

9.4 Reconciliation and correction

Correction is a first-class process, not an afterthought. Three principles govern it. *Correct to truth, forward*. A confirmed error is corrected to the true value in the current computation and propagated forward; the historical record is annotated, not silently edited (Section 3.5). *Never manufacture a fix*. If the true value cannot be reconstructed, the field is marked unresolved and governance is escalated (Section 10.5); a recovery mechanism is never used to mask a fault. *Log everything*. Every correction carries its evidence, its decision, and its version; the correction log is itself auditable.

These principles convert “the system makes errors” — which every measurement system does — into “the system’s errors are bounded, visible, and correctable,” which is the property a fair measurement system must have.

10 Operations and Governance

10.1 Parameter tri-state governance

Every parameter in the system is registered and assigned exactly one of three governance states: *frozen* (permanently settled, changed only by a full method-change process), *initial* (initial-production-effective: the value is the production value, and a calibration study may later propose a change through the method-change process), and *disabled* (retained but not active). The registry is versioned, and every published artifact records the parameter version it was computed under. The tri-state discipline is what makes the V1 weights honest: they are not “placeholder awaiting calibration” — the registry’s *initial* state is production-effective — but the door to calibration is a formal process, not a silent tweak.

10.2 Version vectors

Reproducibility requires that every computation carry its own provenance. The system maintains six version vectors, each incremented when its domain changes, and each published artifact carries the full tuple:

Table 9: Version vectors: each published artifact carries the full tuple, so that “what was computed under what” is mechanical.

Version	Domain	Incremented when
param_ver	hyperparameters	any parameter changes state or value
roster_ver	membership	the included or shadow set changes
method_ver	formulas	any composition formula changes
decision_ver	governance decisions	an adjudicated decision is recorded
flagship_ver	flagship computation	the flagship formula or its inputs version
source_ver	source definitions	a source is added, swapped, or deprecated

Two properties of the tuple matter. *First*, the tuple is monotone in time: versions never decrease, so the tuple is a total order over computations, and “compare two snapshots” is well-defined. *Second*, the tuple is fine-grained enough to localize: a source swap changes `source_ver` without implying a method change, and a reader can in principle see *what kind* of change happened without diffing every formula. The version tuple is the operational form of the paper’s reproducibility posture.

10.3 Settlement calendar and windows

The system settles twice daily in two closed-market windows (UTC 04:00–07:00 and 16:00–19:00), on the same structure and with equal weights for both windows. The anchor is frozen between settlements; the live signal only decays during closed windows (no injection, no renewal), and events that arrive during a closed window are batch-settled at the next opening. Reopenings are announced. The calendar is a governance object: changing the number or placement of windows is a method change with a version entry. The design reasons for two windows are stated in Section 5.7: they bracket the main activity periods of the two hemispheres and bound anchor staleness within a day.

10.4 Dynamic membership operations

Membership changes follow the inclusion rules (Section 3.4) and a stated operational path: a candidate is evaluated at a settlement; if admitted, the admission takes effect at the next settlement and is disclosed with its rationale and its `roster_ver` increment; the no-backcasting discipline (Section 3.5) prevents any retroactive restatement. When an admission changes a normalization extreme, the mechanical revaluation is handled by the BR-01 bridge (Section 6.10), which separates the mechanical from the fundamental component of the change. Shadow status is the intermediate tier: a shadow unit is evaluated and displayed but does not define normalization extremes or enter the centering mean (Section 3.3), which lets the system accumulate evidence about a candidate before the candidate’s admission can move the scale.

10.5 Suspension, hold, and recovery

Integrity events are handled by a graded mechanism. *At the field level*, a suspected bad value is held for re-verification while the most recent valid value continues in use (Section 9.1). *At the settlement level*, if the formal computation cannot be shown to be complete, valid, and reproducible — a P0-class integrity failure — the candidate settlement is held, no new formal version is submitted, and the last confirmed state remains published with a “settlement delayed” notice; the system keeps collecting and recomputing in the background. *At the entity level*, a single Originator may be suspended (frozen intraday and at settlement) on the decision of the arbitration committee, without changing its membership status — suspension is a quotation/governance state, not a membership state. Recovery requires a demonstrated root cause, a provable safety boundary, idempotent recomputation, and passed invariant checks. The graded mechanism is designed so that a fault is isolated at the smallest domain that contains it (the residual risks of Table 7 are the input to this design).

11 FA Case Study: the 2026-08-29 Snapshot

This chapter works through the first FA snapshot as a worked example of the measurement machinery — what it shows, what it does not show, and which of the paper’s claims it evidences. It is *FA-only* by design: the live signal is not reproduced here, and no RI-related number appears. The snapshot is dated (2026-08-29 settlement).

11.1 Snapshot and rank structure

The snapshot table is reproduced in Appendix G (Table 24): twelve Originators ordered by FA, with the two shadow units marked. The table reports the raw anchor FA; at this settlement the sustained-leadership adjustment D is small and $R = 0$, so FA^{adj} differs from FA only marginally and the base-pool conservation holds to within that margin. Three structural features are worth reading off the table.

The leaderboard is close at the top. The top four Originators lie within a band of about eight points, and the top two within one point. The near-tie is reported transparently rather than broken by an arbitrary tiebreaker, per the paper’s boundary disclosure (Section 8.9). A reader should interpret ranks inside this band as “statistically indistinguishable given the current window,” not as a decisive ordering.

The range is compressed at the bottom. The bottom two Originators are the shadow units, and the shadow/included boundary is visible in the data: both shadow units are evaluated with the same parameters but do not define normalization extremes (Section 3.3), so their FA values are computed against the included set’s extremes.

The two component columns are the story. The capability and ecosystem composites (cap_{pts} , eco_{pts}) are deliberately shown alongside the aggregate, because the aggregate alone hides the structure that justifies the composition.

11.2 Capability $\neq$ influence, in data

The strongest single observation in the snapshot is the inversion between capability and ecosystem stock. Two Originators illustrate it sharply.

Anthropic has the highest capability composite in the set yet ranks second-lowest on ecosystem stock among the included units, and its overall FA rank (6) is far below its capability rank (1). *DeepSeek*, by contrast, leads the ecosystem composite while its capability composite is mid-pack, and its overall FA rank (2) is well above its capability rank. These are not measurement artifacts; they are the concrete content of the paper’s premise that capability is necessary but not sufficient for influence (Section 1.2). If the system had measured only capability, it would place Anthropic first and DeepSeek ninth; the ecosystem channel — the “use $\times$ iteration” stock side — moves the ordering substantially.

The inversion is also the discriminant-validity evidence of Section 4.5: the capability and ecosystem channel scores are visibly not a single latent factor (the Spearman correlation across the twelve rows is $\rho \approx 0.007$ — essentially orthogonal, far below any collinearity threshold), which is the precondition for the composition’s weights to mean anything. A composition whose channels are near-collinear would be a disguised single factor; this snapshot shows they are not.

11.3 Missing-value and coverage in practice

The snapshot exercises the missing-value taxonomy in real data. The capability coverage table (Appendix B describes the protocol; the production registry records each classification) shows that no Originator is covered by all seven capability sources; each is covered by a subset, and the composition is computed under two-level renormalization (Proposition 6.4). Three examples illustrate the taxonomy at work:

- *True zero in the edge block.* Several Originators have no model in the parameter bands the edge block measures; their edge-band contribution is zero and *always present in the mean* (Section 6.4). A missing edge band is a fact about participation — the unit does not field small models — and it is counted as such rather than renormalized away.
- *Numerical absence in commerce factors.* An Originator with no commercial routing has no price/value observations; those factors are non-punitively renormalized over its present commerce factors rather than zeroed. Absence of a routing is not evidence of zero commercial accessibility in the way that absence of official free API access is.
- *Leaderboard floor in adoption.* An Originator below the printed cutoff of the adoption board is valued at the last readable rank, taken dynamically from the current board — never hardcoded — so that “not photographed onto the board” is not converted into an absence discount.

The point of the chapter is not that these classifications are the only defensible ones; it is that they are *registered and auditable*. A reader who disagrees with a classification can in principle see which classification was made, why, and trace its effect on the aggregate. That is the difference between a data pipeline and a measurement.

11.4 Conservation in practice

At this settlement $R = 0$ and D is small, so the exact and approximate conservation propositions nearly coincide: the sum of the flagship over the included set equals the sum of the adjusted anchors exactly, and equals the sum of the base anchors up to the size of the D adjustment (Propositions 6.9–6.11). The snapshot therefore doubles as a numeric instance of the conservation law: the published column can be summed and the identity checked to within that margin. This is the partial-reproducibility claim of the paper made concrete — the FA methodology, the reference parameters, and the snapshot are sufficient to check the system’s central invariant, even though the live signal is not reproduced.

11.5 What the case study does and does not show

The snapshot evidences: (i) the composition is computable and bounded (Proposition 6.12); (ii) the channel structure is discriminant in practice; (iii) the conservation invariant holds on released data; and (iv) the missing-value taxonomy is exercised by real coverage gaps. It does *not* evidence: (i) the validity of the index as a measure of influence (that is the running program of Section 4 and Section 8.6); (ii) any dynamic property (one snapshot has no time dimension); or (iii) any claim about the live signal. The boundary is drawn so that the reader does not over-read the snapshot.

12 Outlook and Future Work

12.1 Generality of the FA + RI architecture

The two-component architecture — a slow, stock-like anchor plus a fast, centered flow signal — is not specific to AI. It transfers, with the same conservation semantics, to other influence fields, for example: influence in a scientific community (anchor: publications and citations; flow: research discussion); influence of an open-source ecosystem (anchor: repository and dependency stock; flow: developer attention); influence of a regional or industry ecosystem (anchor: capacity and output stock; flow: news and adoption); and influence of cultural production (anchor: canon and archive stock; flow: public discussion). The object of the current system is AI Originators. The generality claim is stated as a design property (anchor and flow are generic measurement objects), not as a delivered multi-domain system.

12.2 Method roadmap

Three items are planned next. A *unified calibration experiment* will re-estimate the V1 reference parameters in batch against held-out evaluation and report the sensitivity of the invariants to the parameter vector (the design is given in Section 8.7). *Board expansion* will add qualified Originators under the inclusion rules and evolve the inclusion rules themselves as the roster grows. *Source-swap migration drills* will exercise the replaceability mechanism (Appendix D) so that the system’s dependence on any single source is continuously testable. A fourth item runs alongside: the *construct-validity evidence program* of Section 8.6 — collecting the external convergent and event-consequence evidence that only time and independent data can supply.

12.3 Dissemination and public-knowledge infrastructure

The vision of the system couples evaluation methodology with communication science for AI concept and information equality. The index is designed as a piece of public-knowledge infrastructure: the

same artifact that is rigorous enough to be a fairness anchor is simple enough to be read by a layperson (Section 7.4). The system is open to being studied as an experiment in science communication — what an audience does with a single, real-time, conservation-anchored number about AI influence. We do not over-promise: the infrastructure claim is a direction of work, not a delivered outcome, and the reflexive dynamics of a public influence number are themselves a research object (Section 7.5).

12.4 Honest boundaries, restated

The data window is short and will grow; the reference parameters are V1 values whose unified calibration is pending; the live signal is opaque by design and its openness timing is undecided; and the roster is small and will expand. Each of these is a known, stated boundary with a known next step. Two further boundaries are restated because they are the ones most likely to be over-read: the index does not predict (it records what has already happened), and it does not measure influence beyond the extent expressible in verifiable public sources (Section 2.5).

13 Open Data Layer, Sources, and Acknowledgments

13.1 Architecture: method layer decoupled from data layer

EINS is an *assessment-method layer*. It does not produce, own, or sell the underlying data. Its inputs are, as a backbone, third-party open data, and, as a supplement, self-collected public data (e.g., its own observed series for some metrics, marked as such). The correct positioning sentence is: EINS is an assessment method rather than a data product, using third-party open data as its backbone and self-collected public data as a supplement. Because the method layer does not compete with its sources, the system’s sources are not adversaries; they are the foundation of its fairness.

The decoupling has an institutional mechanism, not just a stated intention. The circuit breakers and redundancy of Section 6.2 make any single source replaceable, and the source-swap migration procedure (Appendix D) is a drilled capability. “We do not compete with our sources” is therefore backed by “we cannot be held hostage by any one of them,” which is the actual guarantee of source neutrality.

13.2 FA source acknowledgments

The FA fundamental field is built on the following institutions and source families, and the system gratefully acknowledges them: *Hugging Face* (open model hub and the Open LLM Leaderboard), *GitHub* (open-source repository metrics), *SWE-bench*, *Aider*, and *LiveCodeBench* (coding-capability evaluations), *LMarena* (crowd pairwise comparisons, in the Elo-rating tradition [Elo, 1978, Zheng et al., 2026]), and *OpenAlex* (citation data for the research ripple). The academic literature underpinning the measurement design (Sections 1–2) and the methodological precedents of influence indexing — notably the Stanford AI Index and Epoch AI’s analyses — are cited in the references.

13.3 RI sources: deliberately unnamed

The RI signal consumes “independently verified authoritative media and public platform information,” and the individual sources are *not* named. The reasons are stated plainly: naming media and social-platform sources would (a) create an avoidable legal exposure surface around the collection of public signals and (b) hand an attacker a short-horizon arbitrage map against a component that is already deliberately opaque (Section 7.5). The methodology is described in full at the structural level (Section 6.6); the calibration is not disclosed. Because the paper discloses neither the sources

nor the numerical parameters, the RI description is a description of a mechanism, not a specification that could be gamed; that is the intended epistemic posture of asymmetric disclosure (Section 7.2).

13.4 Publisher attribution

The system is published under a role designation rather than an individual name, for a single reason: fairness of assessment. An evaluation of EINS and the Eins Index should turn on the methodology, the formal structure, the data, and the evidence — not on who published them. Identity-blind assessment is the established device for keeping scrutiny on content rather than provenance; the equity rationale of blind peer review is exactly this [Pleskac et al., 2026], and a measurement system has the same need. The choice withholds only the publisher’s identity: the FA methodology, the inclusion rules, and the reference parameter tables are given in full (Appendices B–C), while the RI layer remains deliberately undisclosed. Accountability lives in the published method, not in a byline. There is also a symmetry to it: a system that measures influence should not mint a personal-influence asset out of its own publisher. The request is therefore simple: judge EINS and the Eins Index on their own merits, and build on the method layer if it earns your confidence.

13.5 Openness commitments

Openness is asymmetric and deliberately modest. The FA methodology and reference parameter tables (Appendices B–C) are documented in this white paper, and FA-related data is intended to be made available to the extent feasible. RI implementation details and the live-signal parameters are reserved. The stated posture is that the system’s *architecture* is open to scrutiny (its invariants are stated publicly and are intended to be checkable), while specific open-scope and open-timing decisions remain under review. Third parties are invited to build on the method layer and to contribute data-layer collaboration. These statements are made without promises of specific scope or timing, because a commitment that outruns its institutional support is itself a failure mode; the architecture is the commitment, and the specific releases follow from it as they become sustainable.

13.6 Disclaimer

EINS is a *research* social-influence measurement system, built for the benefit of public understanding of AI; it is not a commercial project and is not designed for financial use. The Eins Index and its family are research artifacts. They do not constitute financial information, investment advice, a credit assessment, a reputation judgment, or a basis for individual decisions; no action should be taken on the basis of the index without independent verification. The observation window is short and the method is under active iteration. Event attribution and grading include human adjudication, which is subject to revision. Source availability affects coverage. The index reflects what has already happened; it is not a prediction. The live-signal component is deliberately non-reproducible by external parties. The system is an independent third-party evaluation, not an endorsement of any measured Originator, and it does not use the logos of the measured organizations. Legal compliance is addressed in Appendix H. Nothing in this white paper is a promise of any specific index behavior, openness scope, or release timing.

Acknowledgments

The system is grateful to the institutions listed in Section 13.2 for maintaining the open data on which the FA field depends, and to the research community whose evaluation methodology and

influence-measurement literature (Sections 1–2) frame this work.

Appendices

A Symbols, Notation, and the Code–Symbol Binding

A.1 Principal notation

Table 10: Principal notation used in the main text. Numerical values for the FA-related symbols are in Appendix B; RI-related values are not disclosed.

Symbol	Meaning	Where
$\mathcal{U}, \mathcal{I}(t), \mathcal{S}(t)$	universe; included; shadow sets	§6.1
$\xi(t)$	full indicator state	Def. 6.1
$x_{i,k}$	L1 source input of unit i , source k	§6.1
$\hat{v}$	normalized factor value	§6.3
cap, edge, eco, com	FA blocks	(7)–(10)
$\alpha_{\text{cap}}, \alpha_{\text{eco}}$	top-level group weights	(11)–(12)
$\beta_{\text{cap}}, \beta_{\text{edg}}$	within-capability block weights	§6.5
$\beta_{\text{eco}}, \beta_{\text{com}}$	within-ecosystem block weights	§6.5
$\gamma_{\text{dev}}, \gamma_{\text{acad}}$	ecosystem sub-block weights	(9)
$\delta_{\text{rip}}, \delta_{\text{led}}$	academic sub-block weights	§6.5
$\text{FA}_i, \text{FA}_i^{\text{adj}}$	anchor and adjusted anchor	(13), (14)
D_i, R_i	sustained-leadership; real-harm adjustments	(14), (15)
$D_{\text{cap}}, R_{\text{cap}}$	adjustment caps	Appendix B
$\text{streak}_{d,y}, f(\cdot)$	streak length; concave weight function	(15)
w_y, β_d	tier weights; leaderboard weights	(15)
$\text{RI}_i(t), \overline{\text{RI}}(t)$	live signal; included cross-sectional mean	(16), (17)
$A_e, t_e, T_{1/2}(e)$	pulse amplitude; start; half-life	(16)
$\mathcal{E}_i(t)$	active pulse set of unit i	(16), (19)
p	polarity in $[-1, 1]$	§6.6
k_{ri}	live-signal dispersion coefficient (reserved)	(17)
BR	reserved bridge slot	§6.10
$\Theta_i, \tilde{\Theta}_i$	flagship actual; nominal	(1), §6.8
H_i	absolute discussion heat	(19)
Γ_i, Λ_i	capability composite; absolute composite	(20)–(21)
μ_Γ, M_Γ	sector mean; sector maximum of Γ	(23)
$C_i, C(t), C_{\text{frontier}}$	capability level; deflator; frontier constant	(24), §6.7

A.2 The complete code–symbol binding

Product codes are *labels*, *not variables*: they never appear inside formulas. Table 11 is the complete and authoritative binding. The family codes that denote *internal* or *derived* quantities (e.g., 8102) are included; the user-facing publication family is the subset listed in Section 6.7.

The regional composites (CN1111, US1111, CN3333, US3333, CN5555, US5555) are country subsets of the flagship semantics: each is a composite over the country’s Originators ordered by Θ , with the suffix denoting the top- k truncation (1, 3, 5). They share the flagship’s conservation and audit

Table 11: Complete code–symbol binding. Every product code is bound here to the symbol (if any) used in formulas.

Code	Symbol (if any)	Meaning	Defined at
8100	$\Theta_i(t)$	company composite influence, relative (actual)	(1)
8100 (nominal)	$\tilde{\Theta}_i(t)$	same, presentation-layer nominal	§6.8
8101	$\text{FA}_i^{\text{adj}}(t_0)$	fundamental anchor (adjusted), settled	(14)
8102	$\text{RI}_i(t)$	real-time influence, live signal	(16)
8111	$\text{H}_i(t)$	absolute discussion heat	(19)
8200	Γ_i	capability composite (absolute)	(20)
8300	$\Lambda_i(t)$	absolute composite (deflated)	(21)
9200	$\mu_\Gamma(t), M_\Gamma(t)$	sector aggregates (mean / max)	(23)
CN1111	—	China Top-1 composite (by Θ)	§6.7
US1111	—	United States Top-1 composite (by Θ)	§6.7
CN3333	—	China Top-3 composite (by Θ)	§6.7
US3333	—	United States Top-3 composite (by Θ)	§6.7
CN5555	—	China Top-5 composite (by Θ)	§6.7
US5555	—	United States Top-5 composite (by Θ)	§6.7
1111, 3333, 5555	—	regional truncation suffix: Top-1 / Top-3 / Top-5	§6.7

machinery restricted to the country subset, and they exist so that a national audience can read its own landscape without losing the cross-sectional discipline of the full set.

B FA Reference Parameters (V1)

All FA-related parameters below are the V1 production values: calibrated by the expert group and engineering tuning, frozen at release under the tag `fa-v1` with a frozen copy `hyperparams_v1.0.frozen.yaml`, and governed by the registry’s tri-state parameter discipline (frozen / initial-production-effective / disabled). The status column uses “frozen” for permanently settled values and “initial” for values that are initial-production-effective design values of V1; in this registry’s governance, “initial” does *not* mean “to be calibrated later” — the unified calibration experiment (Section 12.2) is a planned sensitivity study, not a prerequisite for these values.

Table 12: FA-1. Trichotomy group and block weights.

Layer	Weight	Status
Capability / ecosystem composite (top level)	0.60/0.40	frozen
Capability block / edge block (within capability)	0.55/0.05	frozen
Ecosystem block / commerce block (within ecosystem)	0.25/0.15	frozen
Developer block / academic block (within ecosystem)	0.50/0.50	frozen
Ripple / ledger (within academic block)	0.50/0.50	frozen

Closed form: with the V1 values, (11)–(13) reduce to $\text{FA}_i = 100 (0.55 \text{ cap}_i + 0.05 \text{ edge}_i + 0.25 \text{ eco}_i + 0.15 \text{ com}_i)$, and the FA value reads as the percentage of the best-weighted combination in the included set.

The blended price is $\text{blend} = (3 \text{ prompt} + 1 \text{ completion})/4$ (3:1 input–output mix); special price tiers (: *batch*, : *contributor*, : *free*) are excluded from value/price.

Within a band, a unit scores its best-model capability divided by the band-wise cross-organization

Table 13: FA-2. Capability block: seven benchmark sources with expert weights.

Benchmark source	Weight	Status
Composite model-capability evaluation	0.40	frozen
Dimensional model-capability evaluation	0.15	frozen
SWE-bench solve rate	0.15	frozen
Aider edit accuracy	0.05	frozen
LiveCodeBench	0.05	frozen
HF Open LLM Leaderboard composite	0.10	frozen
LMarena Elo	0.10	frozen

Table 14: FA-3. Developer block: source weights (heavy-tail scale).

Source	Weight
GitHub stars	0.35
HF downloads	0.30
HF likes	0.15
GitHub repos	0.10
HF models	0.10

maximum over the included set; a missing band contributes 0 and is always present in the mean (true-zero discipline, Section 6.4).

The classification of each factor for each unit is registered and versioned, so that the handling of every missing value is auditable (Section 6.4).

Verification factor $V \in \{0.7 \text{ (public-list)}, 0.9 \text{ (committee)}, 1.0 \text{ (independent)}\}$; the ledger value is $\sum_e T_e V_e$, normalized linearly.

The live-signal coefficient k_{ri} and all RI calibration values are reserved and are not listed.

C First-Assessment Roster and Inclusion

The first-assessment roster is the set of Originators actually carried in the V1 production computation. It consists of the *included set* $\mathcal{I}$ (ten units, all of which enter the flagship’s conservation and centering) and the *shadow set* $\mathcal{S}$ (two units, computed and published for observability but not part of $\mathcal{I}$). The roster was settled in the V1 production freeze; the shadow designation follows the boundary rules of Section 3.4 and is reviewed at the monthly review (Section 10.4).

The shadow designation is not a judgment about influence; it is a statement about the operational completeness of the observation layer for the unit at the review date. Units graduate to $\mathcal{I}$ by the dynamic membership procedure (Algorithm 9 in Appendix F) once the observation layer is judged complete; units exit $\mathcal{I}$ when the layer degrades below the completeness threshold for two consecutive reviews.

D Formal Derivations

D.1 Conservation error of the nominal flagship

The flagship equation (17) is $\Theta_i(t) = \text{FA}_i^{\text{adj}}(t_0) + k_{ri}(\text{RI}_i(t) - \overline{\text{RI}}(t)) + \text{BR}_i(t)$ for $i \in \mathcal{I}$. Summing over the included set and using $\sum_{i \in \mathcal{I}} (\text{RI}_i(t) - \overline{\text{RI}}(t)) = 0$, we obtain the *conservation law in exact*

Table 15: FA-4. Commerce block: four-factor weights and pricing rule.

Factor	Weight	Rule
value	0.20	flagship capability / flagship blended price (same-model pairing)
price	0.20	inverse blended price, heavy-tail
free	0.10	boolean presence of official free API access (true-zero discipline)
momentum	0.50	token adoption, linear direct division, leaderboard floor

Table 16: FA-5. Edge block: parameter-band boundaries (total parameters, billions).

Band	Total parameters (B)
1	(0, 2]
2	(2, 10]
3	(10, 100]
4	(100, 300]
> 300 B: not counted in the edge block.	

form

$$\sum_{i \in \mathcal{I}} \Theta_i(t) = \sum_{i \in \mathcal{I}} \text{FA}_i^{\text{adj}}(t_0) + \sum_{i \in \mathcal{I}} \text{BR}_i(t). \quad (33)$$

The *published nominal* line (Section 6.7) is $\tilde{\Theta}_i(t) = \Theta_i(t) + \tilde{\eta}_i(t)$ (the display layer’s dual-value process, Section 6.8). Because the display noise $\tilde{\eta}_i(t)$ is a balanced, mean-zero process over $\mathcal{I}$ with display variance bounded by σ_0^2 , the nominal flagship satisfies the *approximate* conservation law

$$\left| \sum_{i \in \mathcal{I}} \tilde{\Theta}_i(t) - \sum_{i \in \mathcal{I}} \text{FA}_i^{\text{adj}}(t_0) \right| \leq \left| \sum_{i \in \mathcal{I}} \text{BR}_i(t) \right| + o_{\mathbb{P}}(1), \quad (34)$$

where $o_{\mathbb{P}}(1)$ vanishes as the display average over the included set concentrates around zero. This is the precise sense of “ $\Sigma \Theta \approx \Sigma \text{FA}^{\text{adj}}$ ” stated in the position note and defended in Section 5.6. Note that the adjusted anchor already contains the discrete D -adjustment; the continuous settlement re-anchoring of Section 10.3 restores $\Sigma \Theta = \Sigma \text{FA}^{\text{adj}}$ at each settlement instant (Algorithm 5).

D.2 Renewal dynamics of the live signal

Fix a unit i and, for the derivation, a single event e with amplitude A_e , start time t_e , and half-life $T_{1/2}(e)$. The renewal envelope in (16) contributes $A_e 2^{-(t-t_e)/T_{1/2}(e)}$ for $t \geq t_e$. Write $\rho_e(t) = \ln 2 \cdot (t - t_e)/T_{1/2}(e)$; then the contribution satisfies the ordinary differential equation

$$\frac{d}{dt} \left(A_e 2^{-\rho_e(t)} \right) = -\frac{\ln 2}{T_{1/2}(e)} \left(A_e 2^{-\rho_e(t)} \right), \quad \frac{d}{dt} \left(A_e 2^{-\rho_e(t)} \right) \Big|_{t=t_e^+} = -\frac{\ln 2}{T_{1/2}(e)} A_e, \quad (35)$$

i.e., pure exponential decay with rate constant $\lambda_e = \ln 2/T_{1/2}(e)$ between renewal instants, and an upward jump of size A_e at t_e . The renewal rule of Section 6.6 is exactly the statement that a repeated event with a residual contribution larger than a fresh pulse replaces the residual by the fresh pulse at the *later* start time:

$$A_e 2^{-\rho_e(t)} \leq A_{e'} 2^{-\rho_{e'}(t)} \Rightarrow (\text{renewal at } t_{e'}) : A_e 2^{-\rho_e(t)} \leftarrow 0 \text{ for } t \geq t_{e'}. \quad (36)$$

Table 17: FA-6. Unified scale contract (factor-level).

Class	Formula	Applies to
Linear	$v/\max$	benchmark counts, edge, momentum
Heavy-tail	$\log_{10}(1+v)/\log_{10}(1+\max)$	ecosystem sources, ripple, value
Inverse	$1/v$, then linear or heavy-tail	price
Boolean	0/1 as-is	free

Table 18: FA-7. Missing-value classification (four-state). States P (true zero), Z (numerical absence), A (leaderboard floor), F (technical unavailability) match Section 6.4.

State	Class	Handling
P	True zero (participation with no observable activity)	weight zero, kept present; renormalization forbidden
Z	Numerical absence (source does not cover the unit)	re-normalize over present dimensions
A	Leaderboard floor (below the printed cutoff)	value by the last readable rank, taken dynamically
F	Technical unavailability (source outage / schema failure)	carry most recent valid value forward with alert

Consequently, $\text{RI}_i(t)$ is a right-continuous process with left limits (càdlàg), piecewise exponentially decaying, whose jumps are non-negative, occur at event starts, and are self-limiting through the renewal rule. This structure guarantees: (i) $\text{RI}_i(t)$ is non-negative whenever all initial pulses are non-negative and the renewal rule is monotone; (ii) the heat envelope $\text{H}_i(t)$ (same envelope with $|A_e|$) dominates the signed signal, $|\text{RI}_i(t)| \leq \text{H}_i(t)$ for all t , which is Proposition 6.14; and (iii) the cross-sectional mean $\overline{\text{RI}}(t)$ inherits the renewal properties, so the centered term in (17) is bounded in the sense of Proposition 6.12.

D.3 Source-swap and circuit-breaker migration

The observation layer is decoupled from any single source (Section 6.2). Formally, suppose a factor $x_{i,k}$ was computed from source k with weight w_k , and the circuit breaker switches to a redundant source k' with weight $w_{k'}$ such that $w_k = w_{k'}$ (a *like-for-like swap*) or $w_{k'}$ is the defaulted weight of Table 3. The *factor-level* effect is bounded by the inter-source discrepancy:

$$|\hat{v}(x_{i,k}) - \hat{v}(x_{i,k'})| \leq \Delta_k, \quad \Delta_k := \max_{i \in \mathcal{I}} |\hat{v}(x_{i,k}) - \hat{v}(x_{i,k'})|. \quad (37)$$

Because the scale contract is monotone and bounded in $[0, 1]$ (Section 6.3) and each factor enters the anchor linearly with weight w_k , the anchor-level effect of any single source swap is

$$|\Delta \text{FA}_i^{\text{adj}}| \leq (\text{aggregation gain}) \cdot \sum_{k \in \text{swapped}} w_k \Delta_k, \quad (38)$$

where the aggregation gain is the product of the composition weights along the path from the factor to FA^{adj} (Section 6.5). Since $\Delta_k \leq 1$ and the weights sum to ≤ 1 , the anchor is Lipschitz in the inter-source discrepancy: no single source can move any anchor by more than its own weight times the worst-case discrepancy, and the dual-track audit (Algorithm 7) detects a swap whenever Δ_k exceeds the alert threshold. This is the quantitative form of the “no single source is load-bearing” claim, and it is the basis of the reconciliation test (Algorithm 7).

Table 19: FA-8. Sustained-leadership (D) parameters.

Parameter	Value
Flagship thresholds	$\{1, 2, 3, 5, 10\}$
Edge-side thresholds	$\{1, 3, 5\}$
Weight function	$f(n) = \log_2(1 + n)$
Tier weights w_y	$[1.00, 0.85, 0.70, 0.50, 0.30]$
Leaderboard weights β_d	MCE 0.35; SWB 0.25; LMA 0.20; AID 0.10; HFL 0.10
Cap D_{cap}	+10% of FA (frozen)
Reset	immediate to zero on falling out of all thresholds; no decay tail

Board codes: MCE = model-capability evaluation (Table 13); SWB = SWE-bench; LMA = LMArena; AID = Aider; HFL = the HF Open LLM Leaderboard.

Table 20: FA-9. Adjustment caps.

Cap	Value	Status
D_{cap}	+10% of FA	frozen
R_{cap}	$\leq 5\%$ of FA	initial

D.4 Weight-sensitivity and the Leamer frontier

Let $\text{FA}_i(\beta)$ denote the anchor as a function of the V1 composition weights β (all normalized to sum to one within each group). For a fixed included set, the ranking of two units i, j is unchanged under weight perturbations $\beta \mapsto \beta + \delta$ as long as $\text{FA}_i(\beta + \delta) > \text{FA}_j(\beta + \delta)$. The set of weights that reverse the ordering is a half-space in weight space; its distance from β in the relative metric measures how robust the ordering is. Following the Leamer tradition [Leamer, 1983], define the *reversal radius* of the pair (i, j) as

$$r_{ij} = \inf \left\{ \|\delta\|_1 / \|\beta\|_1 : \text{FA}_i(\beta + \delta) \leq \text{FA}_j(\beta + \delta) \right\}. \quad (39)$$

A ranking is *weight-robust* if r_{ij} exceeds the plausible perturbation scale for all adjacent pairs (i, j) . The sensitivity table in Section 8.7 reports the smallest reversal radius across the board; the Monte-Carlo and extreme-bounds analyses (Algorithm 11) estimate the distribution of r_{ij} under the frozen V1 weights and their uncertainty. This gives a constructive, quantitative acceptance criterion: the board ranking is accepted as robust if the minimum adjacent reversal radius exceeds a stated threshold and the distribution of rank changes over the Monte-Carlo ensemble is contained.

E Proof Corpus

E.1 Propositions

Prop. 1 (Interior membership addition preserves existing normalizations). An admission that does not redefine a normalization extreme leaves every previously included unit’s normalized values — and hence FA — unchanged; the index can grow without “restating” the past.

Prop. 2 (Construct–indicator consistency of the influence definition). The four-channel definition of social influence (use $\times$ discussion $\times$ iteration $\times$ diffusion) maps injectively to the four composition blocks; the operationalization preserves the construct’s meaning.

Table 21: FA-10. Ledger artifact tiers (adjudicated research artifacts).

Tier	Meaning	Value
T_1	record-level	1.0
T_2	milestone-level	2.0
T_3	first-level	3.5
T_4	paradigm-level	6.0
T_5	frontier-opening	10.0

Table 22: FA-11. Presentation-layer display-noise parameters (non-RI). These are display-layer parameters of the nominal line (Section 6.8); they are not RI calibration values.

Parameter	Value
Relative amplitude σ_0	20 bps
Amplitude cap	75 bps
Noise half-life	15 min
Taper window	15 min
Jump probability	0.15% per min per unit
Jump magnitude	$\text{uniform}(3, 5)\sigma$
Display jitter (sub-minute)	$\sigma_j = 10$ bps

Prop. 3 (Circuit-breaker dilution bound). When a degraded source is dropped and the factor is re-normalized over surviving sources, the factor changes by at most $w_{s_0}/(1 - w_{s_0})$, the ratio of the dropped weight to the surviving weight.

Prop. 4 (Robust scale is bounded under a single outlier). The robust scale 1.4826 MAD changes by a bounded amount under a single extreme outlier, whereas the sample standard deviation can increase without bound.

Prop. 5 (Ratio invariance of linear factors). A common proportional change to all raw values leaves every linear-normalized factor — and hence FA — unchanged.

Prop. 6 (Two-level renormalization is a projection). The two-level renormalization of missing values is the orthogonal projection of the full composite onto the subspace of present blocks in the weighted ℓ_2 sense, and stays in $[0, 1]$ when the present block values do.

Prop. 7 (FA monotonicity in its blocks). FA is non-decreasing in every block and in every within-block factor, and strictly increasing in any block with positive weight.

Prop. 8 (Exact conservation of the adjusted pool). At every time, $\sum_{i \in \mathcal{I}} \Theta_i(t) \equiv \sum_{i \in \mathcal{I}} \text{FA}_i^{\text{adj}}(t_0)$; the live term reallocates but never changes the pool total.

Prop. 9 (Approximate conservation of the base pool). The relative gap between $\sum \Theta_i$ and $\sum \text{FA}_i$ is bounded by $D_{\text{cap}} + R_{\text{cap}}$; in the current configuration with no streak and no incident, the sums are equal.

Prop. 10 (Well-posedness and boundedness). Under the unified scale contract and the missing-value taxonomy, every factor and composite lies in $[0, 1]$; hence $\text{FA}_i \in [0, 100]$ for all i .

Prop. 11 (State-pricing consequences). Under the state-pricing axiom and the renewal rule: (a) a renewal jump equals full value minus residual; (b) a fresh event enters at full value with

Table 23: First-assessment roster. The included set has ten units; the shadow set has two. Shadow units are marked with an asterisk.

Unit	Set	Rationale (summary)
Anthropic	$\mathcal{I}$	included
DeepSeek	$\mathcal{I}$	included
Google DeepMind	$\mathcal{I}$	included
Moonshot AI (Kimi)	$\mathcal{I}$	included
Meta AI	$\mathcal{I}$	included
MiniMax	$\mathcal{I}$	included
OpenAI	$\mathcal{I}$	included
Qwen (Alibaba)	$\mathcal{I}$	included
xAI	$\mathcal{I}$	included
Zhipu AI	$\mathcal{I}$	included
Mistral AI	$\mathcal{S}$	shadow
01.AI	$\mathcal{S}$	shadow

zero response lag; (c) at most one active pulse exists per primary key and the renewal log is a deterministic replay ledger.

Prop. 12 (Audit inequality). For every Originator and time, $H_i(t) \geq |RI_i(t)|$ over the same envelope set — the heat line is an upper bound on the signed signal.

Prop. 13 (Monotonicity of sector aggregates, conditional). Holding membership, scale, and method fixed, if every member’s Γ is non-decreasing then the sector mean and maximum are non-decreasing.

Prop. 14 (Rank invariance under the deflator). Multiplication by a common positive deflator preserves cross-sectional ranking and relative gaps of Θ at every time.

Prop. 15 (Nominal display is drift-free and converges). The display-noise process has zero mean and is forced to zero at the close of each window; the nominal line does not drift from the actual line and equals it at the close.

Prop. 16 (Centering preserves cross-sectional ordering of the anchor). With equal live signals, flagship ordering is determined entirely by anchor ordering; the live term cannot flip it.

Prop. 17 (BR transition convergence). Under (32): the bridge is zero at time H and thereafter; the published path converges to the new fair value; the bridge never deposits into FA and never enters the RI feed; with no mechanical change no bridge is generated.

E.2 Lemmas, corollary, and property

Lemma A (Decay kernel positivity). For $t \geq t_e$ and $T_{1/2}(e) > 0$, the kernel $2^{-(t-t_e)/T_{1/2}(e)}$ lies in $(0, 1]$ and is strictly decreasing in t .

Lemma B (Renewal jump magnitude). A single renewal jump of pulse e equals $A_e - r_e(t)$, the new full value minus the residual of the voided pulse at the renewal time; the jump is therefore non-negative.

Corollary A (Zero-sum character of the live term). The live term $k_{\text{ri}}(\text{RI}_i - \overline{\text{RI}})$ has zero sum over the included set at every time; it reallocates within the pool and never changes the pool total.

Property A (Zero-reset of sustained leadership). Dropping out of all leadership thresholds sets $D = 0$ immediately with no decay tail; re-entry accumulates from the current tier. No double-counting of inertia that the stock dimensions already carry.

F Algorithms

Each algorithm is stated against the formal framework of Section 6 and is an executable contract of the computation pipeline. The algorithms are versioned through the method version vector (Section 10.2); a change to any of them increments `method_ver` and is logged in the audit trail.

Algorithm 1 Layered observation-state computation

```

1: Input: raw source records  $r \in \mathcal{R}$ ; roster  $\mathcal{I} \cup \mathcal{S}$ ; scale contract  $\mathcal{S}$ ; source registry  $\mathcal{K}$ .
2: procedure L1_INGEST( $\mathcal{R}$ )
3:   for each record  $r$  do
4:     classify key (org, source, title);
5:     stamp arrival time; validate schema against source profile
6:     assign source state: present / degraded / absent (Section 6.2)
7:   end for
8:   return raw records with source states
9: end procedure
10: procedure L2_DERIVE(raw,  $\mathcal{S}$ )
11:   classify missingness per factor into  $P/Z/A/F$  (Algorithm 2)
12:   apply scale contract per factor class (linear / heavy-tail / inverse / boolean)
13:   aggregate factors to blocks by composition weights (Algorithm 3)
14:   return derived indicator state  $\xi(t)$  (Definition 6.1)
15: end procedure
16: procedure L3_PUBLISH( $\xi(t)$ , roster)
17:   compute  $\text{FA}^{\text{adj}}$ , RI envelope,  $\Theta$ ,  $\Gamma$ ,  $\Lambda$ 
18:   apply settlement re-centering and presentation noise (Algorithms 5, 6)
19:   return published layer values
20: end procedure

```

Together, Algorithms 1–11 cover the full pipeline from raw observation to published value, with the verification and governance loops designed to keep the pipeline auditable. The version-vector bookkeeping (Section 10.2) tags every algorithm run, so that a state change can in principle be replayed from the recorded inputs.

Algorithm 2 Missing-value classification and two-level renormalization

```
1: Input: factor values  $v_{i,k}$ ; block weights  $w_k$ ,  $\sum_k w_k = 1$ ; status registry  $\mathcal{M}$ .
2: procedure CLASSIFY( $i, k$ )
3:   if unit not covered by the factor's definition then return  $P$  (true zero)
4:   else if source state absent and no readable rank then return  $Z$  (numerical absence)
5:   else if below printed cutoff, last readable rank available then return  $A$  (leaderboard floor)
6:   else if source outage / schema failure then return  $F$  (technical unavailability)
7:   end if
8: end procedure
9: procedure RENORMALIZE( $i$ )
10:   $\mathcal{P} \leftarrow \{k : \text{state}(i, k) \in \{P, A, Z\text{-with-fill}\}\}$ 
11:   $v_{i,k} \leftarrow 0$  for  $k$  in state  $Z$  without fill, or floor value for  $A$ 
12:   $V_i \leftarrow \sum_{k \in \mathcal{P}} w_k \hat{v}_{i,k} / \sum_{k \in \mathcal{P}} w_k$ 
13:   $\text{carry}(i, k) \leftarrow v_{i,k}(t_{\text{last-valid}})$  for state  $F$ 
14:  register  $(i, k, \text{state})$  in  $\mathcal{M}$  with version stamp
15:  return  $V_i$  (projection onto present blocks, Prop. 6.4)
16: end procedure
```

Algorithm 3 Fundamental-Anchor composition (V1)

```
1: Input: normalized blocks cap, edge, eco, com  $\in [0, 1]$ ; V1 weights (Appendix B); roster.
2: procedure COMPOSE(cap, edge, eco, com)
3:   $\text{cap}' \leftarrow (0.55 \text{cap} + 0.05 \text{edge}) / 0.60$ 
4:   $\text{eco}' \leftarrow (0.25 \text{eco} + 0.15 \text{com}) / 0.40$ 
5:   $\text{FA}_i \leftarrow 100 \times (0.60 \text{cap}' + 0.40 \text{eco}')$ 
6:  merge per block from its factors by the composition map
7:  return  $\text{FA}_i$ 
8: end procedure
9: procedure ADJUST( $\text{FA}_i$ , streaks, incidents)
10:   $D_i \leftarrow$  sustained-leadership value ((15)), capped at  $D_{\text{cap}}$ 
11:   $R_i \leftarrow$  real-harm value, capped at  $R_{\text{cap}} \text{FA}_i$ , official facts only
12:   $\text{FA}_i^{\text{adj}} \leftarrow \text{FA}_i (1 + D_i) - R_i$ 
13:  freeze  $\text{FA}_i^{\text{adj}}(t_0)$  at settlement; immutable intraday
14:  return  $\text{FA}_i^{\text{adj}}(t_0)$ 
15: end procedure
```

Algorithm 4 Real-time-influence renewal update

```
1: Input: event stream; residual map  $r(\cdot)$ ; half-life map  $T_{1/2}(\cdot)$ .
2: for each event  $e$  at time  $t_e$  do
3:    $r_{\text{new}} \leftarrow A_e 2^{-0}$  (full value)
4:   if  $r(\text{key}) \leq r_{\text{new}}$  then ▷ renewal rule
5:     void residual of the same primary key
6:     insert  $r_{\text{new}}$  with start  $t_e$ 
7:     append to the renewal log (deterministic replay ledger)
8:   else
9:     keep existing pulse; log the rejected event
10:  end if
11: end for
12: for each active pulse do
13:    $\rho_e \leftarrow (\text{now} - t_e)/T_{1/2}(e)$ ; value  $\leftarrow A_e 2^{-\rho_e}$ 
14: end for
15:  $\text{RI}_i(t) \leftarrow \sum_{\text{active } e} A_e 2^{-(t-t_e)/T_{1/2}(e)}$ 
16:  $\text{H}_i(t) \leftarrow \sum_{\text{active } e} |A_e| 2^{-(t-t_e)/T_{1/2}(e)}$ 
17: return  $\text{RI}_i(t)$ ,  $\text{H}_i(t)$  (Prop. 6.14)
```

Algorithm 5 Settlement and centering

```
1: Input: fresh  $\text{FA}_i^{\text{adj}}(t_0)$  recomputed from settled slow data ((14)),  $\text{RI}_i$ , roster  $\mathcal{I}$ .
2:  $t \leftarrow$  window close (UTC 04:00–07:00, 16:00–19:00)
3: procedure SETTLE
4:    $\overline{\text{RI}} \leftarrow \frac{1}{|\mathcal{I}|} \sum_{j \in \mathcal{I}} \text{RI}_j(t)$ 
5:   for  $i \in \mathcal{I}$  do
6:      $\Theta_i(t) \leftarrow \text{FA}_i^{\text{adj}}(t_0) + k_{\text{ri}}(\text{RI}_i(t) - \overline{\text{RI}}) + \text{BR}_i(t)$ 
7:   end for
8:   ▷ exact conservation:  $\sum \Theta_i = \sum \text{FA}_i^{\text{adj}}$  (Prop. 6.9)
9:   freeze  $\text{FA}_i^{\text{adj}}(t_0)$  as the new anchor ▷ slow anchor, not the closing line
10:  write settlement record; increment flagship_ver
11:  return  $\Theta_i(t)$ , frozen  $\text{FA}_i^{\text{adj}}(t_0)$ 
12: end procedure
```

Algorithm 6 Presentation-layer nominal dual-value process

```
1: Input: actual line  $\Theta_i(t)$ ; display parameters (Table 22).
2: procedure NOMINALIZE( $\Theta_i(t)$ )
3:    $x \leftarrow 0$  (display-noise stock, hash-seeded per unit)
4:   for each publish tick do
5:      $x \leftarrow x \cdot (1 - 2^{-1/15}) + \mathcal{N}(0, \sigma_0^2)$  ▷ discrete OU, zero mean
6:     with prob. 0.15% per minute: jump  $x \leftarrow x \pm U(3, 5) \sigma$ 
7:      $x \leftarrow \text{clip}(x, \pm 75 \text{ bps})$ 
8:     if inside taper window:  $\tau$  min to close then
9:        $x \leftarrow x \cdot (1 - \tau/N_{\text{taper}})$  ▷ force to zero at close
10:    end if
11:     $\tilde{\Theta}_i(t) \leftarrow \Theta_i(t) + x + \mathcal{N}(0, \sigma_j^2)$ 
12:  end for
13:  return  $\tilde{\Theta}_i(t)$ ; at close  $\tilde{\Theta} \equiv \Theta$  (Prop. 6.17)
14: end procedure
```

Algorithm 7 Dual-track audit and replay

```
1: Input: production trace; replay engine; keys.
2: procedure AUDIT(window  $W$ )
3:   recompute all outputs from raw inputs over  $W$  (replay path)
4:   compare replay vs. production at each published tick
5:   check  $H_i(t) \geq |RI_i(t)|$  over  $W$  (Prop. 6.14)
6:   check conservation  $|\sum \Theta_i - \sum FA_i^{\text{adj}}| \leq \epsilon$  at each settlement
7:   verify renewal-log determinism (re-run key events)
8:   return audit verdict + exception list
9: end procedure
10: procedure RECONCILE(exception list)
11:   for each exception do
12:     classify: display noise / source swap / model change / operator error
13:     if source-related: trigger Algorithm 8
14:     emit correction record; increment decision_ver
15:   end for
16: end procedure
```

Algorithm 8 Circuit breaker and source-swap migration

```
1: Input: source state; factor weights  $w_s$ ; alert threshold.
2: procedure MONITOR( $k \in \mathcal{K}$ )
3:   if health( $k$ ) below threshold for 2 consecutive windows then
4:     degrade source  $k$ : drop its observations, re-normalize over survivors
5:     assert  $|\hat{v} - \hat{v}_{\text{out}}| \leq w_k/(1 - w_k)$  (Prop. 6.1)
6:     mark factor state  $F$  (technical unavailability), carry forward (Section 6.2)
7:   end if
8:   if  $\Delta_k >$  alert threshold on recovery then
9:     record discrepancy; report to reconcile (Algorithm 7)
10:  end if
11:  return circuit state; update source_ver
12: end procedure
```

Algorithm 9 Dynamic membership inclusion

```
1: Input: universe  $\mathcal{U}$ ; observation-layer completeness  $c_u$ ; threshold  $c^*$ .
2: procedure REVIEW( $u \in \mathcal{U}$ )
3:   compute  $c_u$  = fraction of required factors observed over last  $N$  windows
4:   if  $u \in \mathcal{S}$  and  $c_u \geq c^*$  for 2 consecutive reviews then
5:     move  $u$  to  $\mathcal{I}$ ; apply Prop. 3.1 check
6:     if a normalization extreme changes: disclose revaluation, no backcast
7:   else if  $u \in \mathcal{I}$  and  $c_u < c^*$  for 2 consecutive reviews then
8:     move  $u$  to  $\mathcal{S}$ ; re-run centering on the new  $\mathcal{I}$ 
9:   end if
10:  return roster; increment roster_ver
11: end procedure
```

Algorithm 10 BR-01 bridge transition

```
1: Input: mechanical-change gap  $G_{\text{mech}}$ ; horizon  $H$ ; path  $\varphi(t)$ .
2: procedure TRANSITION
3:   if  $G_{\text{mech}} = 0$  then
4:     return (no bridge; Prop. 6.19(d))
5:   end if
6:    $\text{BR}(0) \leftarrow -G_{\text{mech}}$ 
7:    $I^{\text{pub}}(0) \leftarrow I^+ + \text{BR}(0) = I^- + \Delta_{\text{fund}}$  (value preserved)
8:   for  $t \in (0, H]$  do
9:      $\text{BR}(t) \leftarrow -G_{\text{mech}} \varphi(t)$ ,  $\varphi(0) = 1$ ,  $\varphi(H) = 0$ 
10:     $I^{\text{pub}}(t) \leftarrow I^+ + \text{BR}(t)$ 
11:  end for
12:   $\text{BR}(t) \leftarrow 0$  for  $t > H$ ; bridge never deposits into FA, never enters RI
13:  return published path; increment method_ver
14: end procedure
```

Algorithm 11 Weight-sensitivity analysis

```
1: Input: frozen V1 weights  $\beta$ ; factor uncertainty model  $\Theta_{\text{unc}}$ ;  $M$  draws.
2: procedure ANALYZE
3:   for  $m = 1, \dots, M$  do
4:      $\beta_m \leftarrow \beta \cdot (1 + \varepsilon_m)$ ,  $\varepsilon_m \sim$  bounded noise per block
5:     recompute all FA,  $\Theta$ ; record full board ranking  $\pi_m$ 
6:     for each adjacent pair  $(i, j)$  do
7:        $r_{ij} \leftarrow$  smallest relative perturbation reversing the ordering (Appendix D.4)
8:     end for
9:   end for
10:  report: min adjacent  $r_{ij}$ ; distribution of rank changes; extreme-bounds band
11:  return sensitivity report (Section 8.7)
12: end procedure
```

G FA Snapshot

This table reproduces the FA snapshot of the case study (Section 11, specifically Section 11.1). It is the value at the 2026-08-29 settlement, verified computationally from the V1 production inputs; it is a *snapshot*, not a time series. The table reports the raw anchor FA; at this settlement the sustained-leadership adjustment D is small — at most a few percent of FA — and no confirmed incident is recorded ($R = 0$), so the adjusted anchor differs from the raw anchor only marginally. The two shadow units are marked with an asterisk; the ten included units define the normalization extremes and the centering mean. The full code–symbol binding for the published family is in Appendix A (Table 11).

Table 24: FA snapshot, 2026-08-29 settlement (V1 production values, verified computationally). The cap_{pts} , eco_{pts} columns are the two composite channels expressed in FA points ($\text{cap}_{\text{pts}} = 60 \text{cap}^*$, $\text{eco}_{\text{pts}} = 40 \text{eco}^*$) and sum to the FA column up to display rounding. Units marked * are shadow.

Rank	Originator	cap_{pts}	eco_{pts}	FA
1	Google DeepMind	49.89	27.26	77.15
2	DeepSeek	42.16	34.40	76.56
3	Qwen (Alibaba)	48.62	21.12	69.74
4	OpenAI	50.39	18.57	68.97
5	Zhipu AI	45.58	21.33	66.91
6	Anthropic	53.86	11.61	65.47
7	xAI	49.82	10.78	60.60
8	Moonshot AI (Kimi)	46.47	13.03	59.50
9	Meta AI	43.51	15.96	59.47
10	MiniMax	36.55	16.82	53.37
11	Mistral AI *	35.15	16.70	51.85
12	01.AI *	40.12	8.32	48.44

First, the top of the board is close: the top four units lie within a band of about eight points and the top two within one point; ranks inside that band should be read as “indistinguishable given the current window,” per the boundary disclosure (Section 8.9). Second, the shadow units are computed with the same parameters but against the included set’s extremes, so their values are comparable in method but not in set-membership (Section 3.3). Third, the conservation law has an empirical instance here: with D small and $R = 0$, Proposition 6.11 bounds the gap between $\sum_{i \in \mathcal{I}} \Theta_i$ and $\sum_{i \in \mathcal{I}} \text{FA}_i$ by the size of the adjustment, so the table can be read as a near-conservative reallocation of the pool.

Fourth, the balance reading of Remark 6.2 has a concrete instance. The block column sums over the included set are $S_{\text{cap}} = 8.29$, $S_{\text{edge}} = 2.15$, $S_{\text{eco}} = 5.83$, $S_{\text{com}} = 3.01$, so the pool is $\sum_{i \in \mathcal{I}} \text{FA}_i = 657.7$, i.e. 65.8% of the maximal pool 1000 (Table 25). Among the capability benchmarks the extremes bound the spread: LMA-01 (crowd Elo) has column sum 9.83, the most evenly spread source — every unit near the crowd-Elo leader — whereas HFL-01 (open-source downloads/likes) has 3.64, the most concentrated within the block, covered by only four of the ten included units.

G.1 Block-level plausibility audit

The formal validity of the composite is established by the propositions (Section 6.9); this subsection adds a plausibility audit at block level: for each of the four blocks the leader’s identity should line up

Table 25: FA pool and block column sums over the included set ($N = 10$); shares are of the maximal block sum 10 and the maximal pool 1000.

Quantity	Value	Share of maximum
Capability (cap)	8.29	82.9%
Edge (edge)	2.15	21.5%
Ecosystem (eco)	5.83	58.3%
Commerce (com)	3.01	30.1%
Pool ($\sum_{i \in \mathcal{I}} \text{FA}_i$)	657.7	65.8%

with a defensible, independently checkable real-world position, and a systematic coincidence across all four blocks would be unlikely if the composite were measuring something other than influence stock.

Table 26: Block-level plausibility audit: for each of the four blocks, the leader, its normalized value, and the observable basis for the leadership. Values are normalized over the included set $\mathcal{I}$ (Section 6.5); a value of 1.0 means the unit is maximal on every source entering the block.

Block	Leader	value	observable basis
cap	Anthropic	0.979	frontier on reasoning and coding evals; leads four of the seven capability sources, while OpenAI leads two and Qwen leads one — no single unit owns all sources, consistent with $0.979 < 1$
edge	Qwen	1.000	the only unit with a best-in-class model in all four parameter bands; the edge weight rewards per-band leadership explicitly (+5.0 FA points, visible in the cap_{pts} column)
eco	Google DeepMind	0.874	leading adjudicated-ledger academic impact (250,994 ripple citations; ledger $T \times V$ 41.5 vs. 34.45), with open releases (Gemma) lifting the developer side; DeepSeek leads the developer sub-block
com	DeepSeek	0.900	best price–quality blend (raw value 592 vs. next-best 118.7; blend price 0.0875 USD) and the widest usage momentum (20.9T tokens)

Capability block. The capability block cap_i is led by Anthropic at 0.979. The lead rests on dispersed sources rather than a single benchmark: Anthropic leads four of the seven capability sources (the two model-capability-evaluation indices, SWE-bench, and LMArena Elo), while OpenAI leads two and Qwen leads one. No single unit owns all seven sources, consistent with a leader value of 0.979 rather than 1.000; the multi-source structure means the block’s reading is not hostage to any one benchmark.

Edge block. The edge block edge_i is led by Qwen at 1.000 — the only unit with a best-in-class model in every one of the four parameter bands ($s_{i,c} = 1$ in each). This is an explicit reward for edge competence: with the edge weight in (13) at 0.05 and $\text{edge}_i = 1.0$, the block contributes 5.0 FA points, visible in the cap_{pts} column of Table 24. Units with no model in any measured band — Anthropic, MiniMax, xAI, Zhipu — contribute zero by design: a missing band is a true zero about

participation, not a discount.

Ecosystem block. The ecosystem block eco_i is led by Google DeepMind at 0.874 on the academic side: its academic sub-block is 1.000 (leading on both ripple citations and the adjudicated ledger), its developer side is lifted by open releases (Gemma), and DeepSeek leads the developer sub-block, first on all five developer sources. The block is not a blanket closed-source penalty; it rewards open engagement and academic impact jointly. OpenAI (0.611) is carried by academic impact despite being closed, while Anthropic (0.398) and xAI (0.349) sit at the bottom for different reasons: xAI’s academic sub-block is near zero (0.008) despite a strong developer sub-block, and Anthropic’s open-release presence is thin, with only moderate developer and academic sub-blocks.

Commerce block. The commerce block com_i is led by DeepSeek at 0.900 on the strongest price-quality blend in the set (raw value 592 against a next-best 118.7, at a blend price of 0.0875 USD) and the widest usage momentum (20.9T tokens); its only zero factor is free, so the block value is 0.900 rather than 1.000.

The four block leaders are four different units — Anthropic (capability), Qwen (edge), Google DeepMind (ecosystem), DeepSeek (commerce) — exactly what one expects from a composite that is not a proxy for any single company. Block-level values and sub-blocks are published in full, so the snapshot is transparent at block level, while the timing signal RI remains undisclosed per the disclosure posture (Section 7.2; Appendix H).

H Compliance Notes

This appendix records the compliance posture of the system. It complements, and does not replace, the disclaimer of Section 13.6.

H.1 Data, legal, and source governance

The system is a *research-oriented social-influence evaluation system*, not a commercial financial-information or rating service. The framing of the public artifacts — “real-time sequence,” “periodic review-and-release,” “pause and resume” — is chosen to describe a measurement regime, not a securities quotation or a credit-rating opinion. The disclaimer (Section 13.6) states explicitly that the published values are not financial information, not investment advice, not a credit rating, not a reputation judgment, and not a basis for individual decisions. The pricing of this posture is that the system renounces the “market quotation” framing entirely.

Public-information processing. The FA input layer consumes public, non-personal or de-aggregated information from third-party open platforms. The general privacy regime for processing already-public personal information in reasonable scope is relied upon, with three operational obligations honored: (i) a refusal and correction channel for any data subject whose information enters an aggregate, (ii) aggregation before publication so that no individual decision is made from any single record, and (iii) no transfer or resale of personal information. The GitHub terms of use, which permit research use of public, non-personal repository metadata while prohibiting the sale of personal information or the resale of the platform as a service, were reviewed and the system’s use of repository counts, stars, and forks conforms to them; the arXiv API, which is open and requires acknowledgment, is used for bibliographic verification and is acknowledged in Section 13.2. Platform terms change; this record is reviewed at each parameter governance review (Section 10.1).

Reputation and trademark. The real-harm adjustment R is fed only by *officially confirmed* incidents, each with a source and a correction channel, and is capped at R_{cap} (Appendix B); the system does not adjudicate contested facts. The system does not use product logos, and public artifacts carry a statement that the evaluation is that of an independent third-party system and is not an endorsement of any organization.

Source governance and openness. The “sources are friends” (Section 13.1) posture and the circuit-breaker redundancy (Section 6.2) mean the system is replaceable at every source and does not sell, block, or privilege any underlying data. The openness posture is deliberately modest: the system aims to describe the method and to keep hard-to-manipulate artifacts observable over time, without promising a specific open scope or a specific release schedule (Section 13.5).

The residual compliance risks that remain under active review are the precise legal classification of the publication language in some jurisdictions and the exact terms of certain platform data policies that could not be re-verified at writing. Neither is a hard obstacle to the research use described here; both are tracked in the governance log.

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